

Topic : Graphs

Part 1

Graphs

Section 1	Introduction
Section 2	Graphs
Section 3	Walks, Paths and Circuits

Part 2

Trees

Section 1	Trees
Section 2	Rooted Trees
Section 3	Spanning Trees

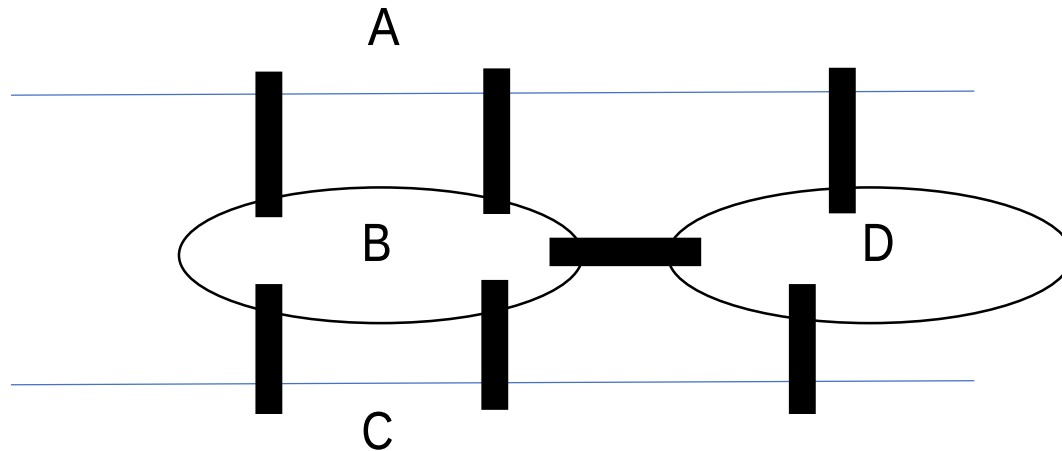
Part 1 Graphs

Section 1 Introduction

- How Did Graph Theory Come About?
- Scope and Objective

How Did Graph Theory Come About?

- In 1736, in the old city of Königsberg in Prussia, there was a River Pregel which flowed through its center.
- There were seven bridges (in black) over the river connecting the two islands (B and D) and two opposite banks (A and C) as shown below :

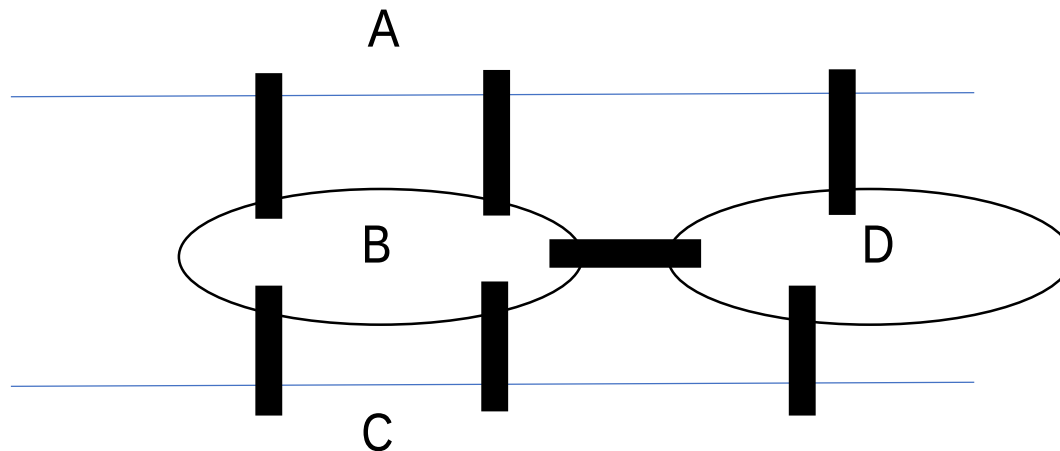


How Did Graph Theory Come About?

People in the city had always amused themselves with this problem :

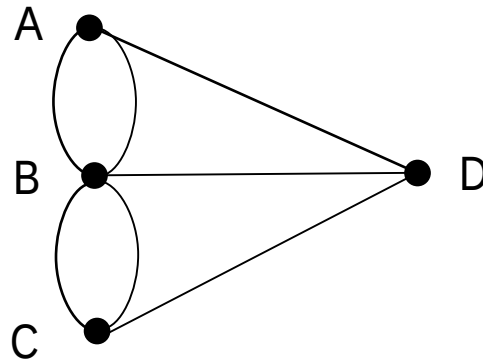
- “Starting from any one of the four places A, B, C or D,
- is it possible to have a walk which passes through each of the seven bridges once and only once,
- and return to where you started? “

Try this!



How Did Graph Theory Come About?

- Euler (pronounced as Oy-ler), the person who solved this problem, translated this into a graph theory problem.
- He represented each island/bank as a dot and each bridge as a line.
- The problem can then be represented as follows :



- So the question now becomes : Is it possible to trace this graph, starting and ending at the same point, without lifting your pencil?
- The answer to this question is : It is impossible to start and end at the same place and yet cross each bridge exactly one time only.

Scope and Objective

- In this course, we will not be able to delve deep enough into Graph Theory to provide a formal proof for the previous question.
- However, we will learn that graphs can be convenient visualizations to model a variety of situations.
- The objective is therefore to learn the following :
 - Terminologies in Graphs
 - To understand some of the concepts of Graphs
 - Common applications of Graphs

Part 1 Graphs

Section 2 Graphs

- Introduction
- Terminologies
- Drawing of Graphs
- Applications of Graphs

Introduction

A graph G consists of two sets :

- a set $V(G)$ of vertices, and
- a set $E(G)$ of edges,

where each edge is associated with either one or two vertices.

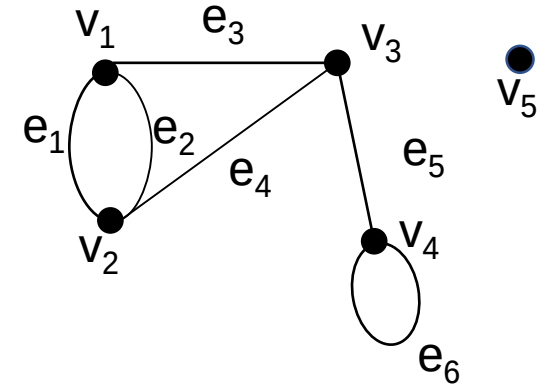
Terminologies of Graphs

Vertex Set

- The set of vertices are the dots in the graph :

$$V(G) = \{v_1, v_2, v_3, \dots v_m\}$$

A vertex with no edges is called an isolated vertex.



Edge Set

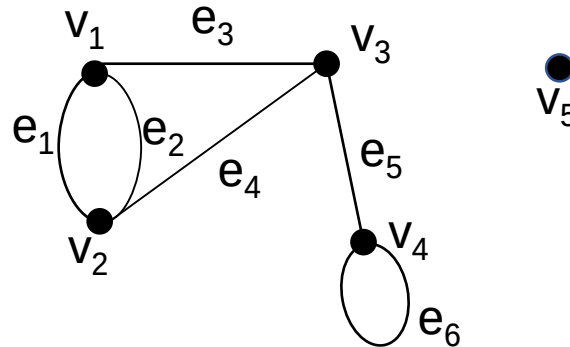
- The set of edges are the lines and curves in the graph :

$$E(G) = \{e_1, e_2, e_3, \dots e_n\}$$

- Each edge can be represented as an ordered pair : $e_x = (v_i, v_j)$
- The order of v_i and v_j within the ordered pair does not matter.
- A loop starts and end at the same vertex. So $e_x = (v_i, v_i)$.
- Parallel edges start and end at the same two vertices, so can be written as (v_i, v_j) and (v_j, v_i) or as repeated (v_i, v_j) .

Example (Terminologies of Graphs)

Consider the following graph G :



- Write the vertex set, $V(G)$, and the edge set, $E(G)$.
- Identify a set of parallel edges, a loop and an isolated vertex.

Solution :

a. $V(G) = \{v_1, v_2, v_3, v_4, v_5\}$

$$E(G) = \{e_1, e_2, e_3, e_4, e_5, e_6\}$$

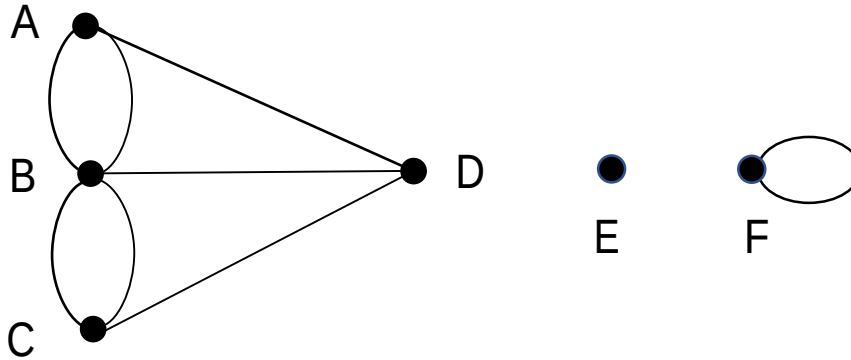
b. Parallel : e_1 and $e_2 = (v_1, v_2)$ and (v_2, v_1)

Loop : (v_4, v_4)

Isolated vertex : v_5

Exercise (Terminologies of Graphs)

Consider the following graph G :



- Write the vertex set, $V(G)$, and the edge set, $E(G)$.
- Identify a set of parallel edges, a loop and an isolated vertex.

Solution :

a. $V(G) = \{A, B, C, D, E, F\}$

$$E(G) = \{(A,B), (B,A), (B,C), (C,B), (A,D), (B,D), (C,D), (F,F)\}$$

b. Parallel : (A,B) and (B,A) or (B,C) and (C,B)

Loop : (F,F)

Isolated vertex : E

Example (Drawing Graphs)

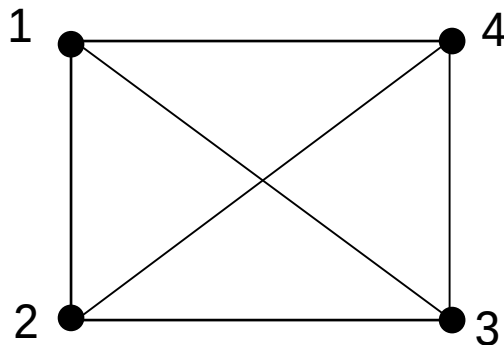
Given the following vertices and edges,

- $V(G) = \{1, 2, 3, 4\}$
- $E(G) = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$

draw a pictorial representation of this graph.

Solution :

After piecing them together, we will have the following graph :



Exercise (Drawing Graphs)

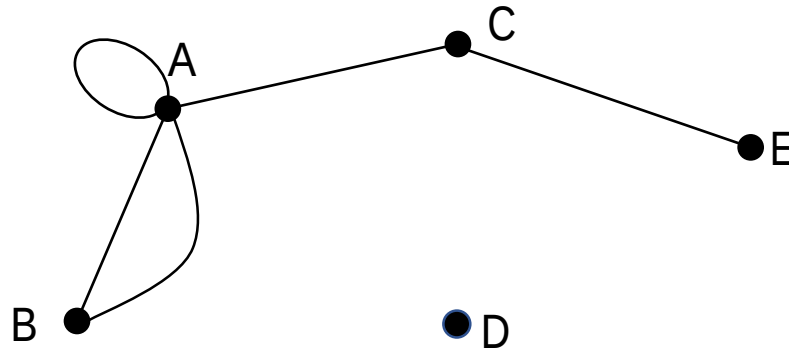
Given the following vertices and edges,

- $V(G) = \{A, B, C, D, E\}$
- $E(G) = \{(B,A), (C,E), (C,A), (A,A), (A,B)\}$

draw a pictorial representation of this graph.

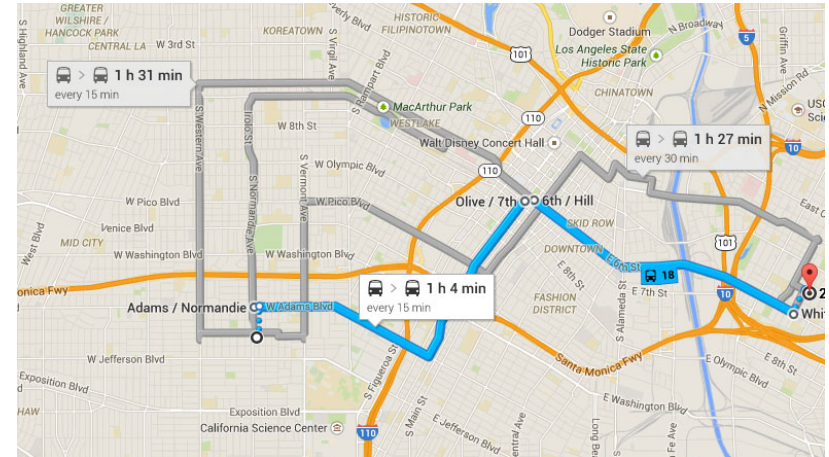
Solution :

After piecing them together, we will have the following graph :

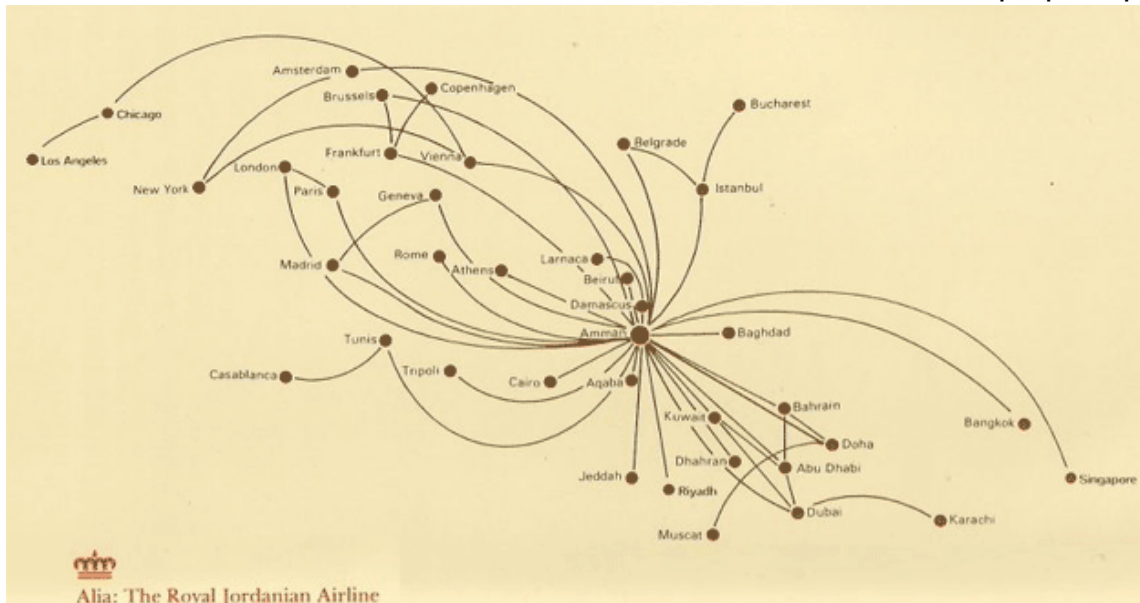


Application : Communication Systems

- Computer networks, internet, telephone network, electric power, gas pipeline, transport systems, social media network etc.
- Basically we are looking for edges of the graphs that minimize time / cost or optimize resources.



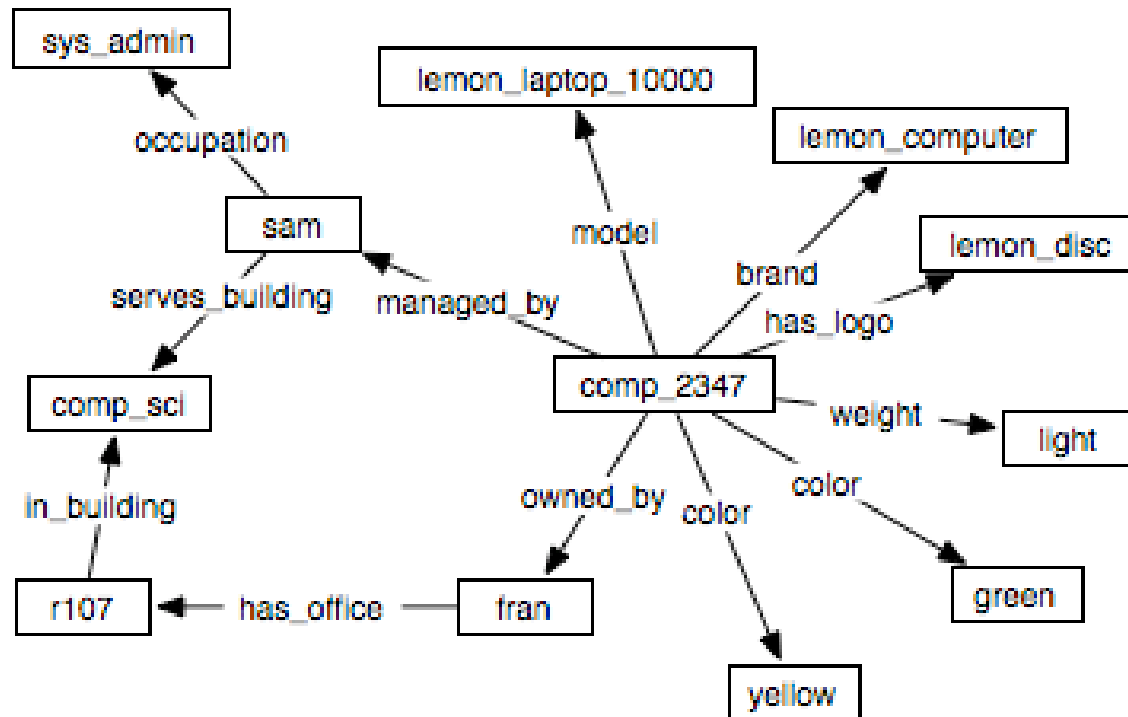
<http://stackoverflow.com/questions/27657321/google-map-api-display-of-alternative-bus-routes-possible>



<https://bluesyemre.com/2013/04/30/4-ways-to-improve-school-communication-using-social-media/>

Application : Knowledge in Artificial Intelligence

- In artificial intelligence, a set of knowledge is represented as a graph and input into the computer. It is called a semantic network.
- Basically, we want the artificial intelligence program to make inference based on the graph.
- What can you infer from the following graph about Sam and Fran?



Part 1 Graphs

Section 2

Walks, Paths and Circuits

- Introduction
- Walks, Paths & Circuits
- Connected Graphs

Introduction

- One of the uses of graphs is to travel through them so as to find the route that minimizes time/cost or optimizes resources.
- Graph travel is accomplished by moving from one vertex to another along a sequence of adjacent edges.
- Certain sequences of adjacent vertices and edges are of importance
 - Those without repeated edge
 - Those that start and end at the same vertex.
- In this course, the focus will be on the following graph travels:
 - Walk, Path and Circuit;
 - Connected Graphs.

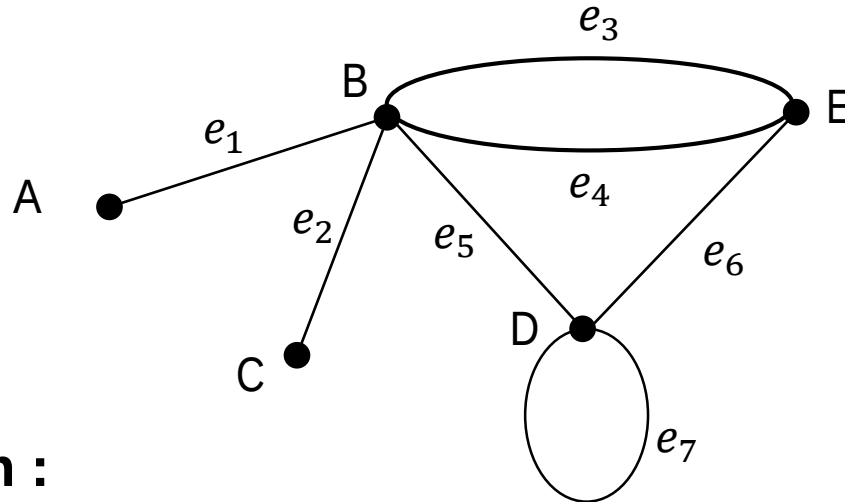
Definition of a Walk

- The first type of graph travel is a walk.

A walk from a vertex u to a vertex v is an alternating sequence of adjacent vertices and edges of G .

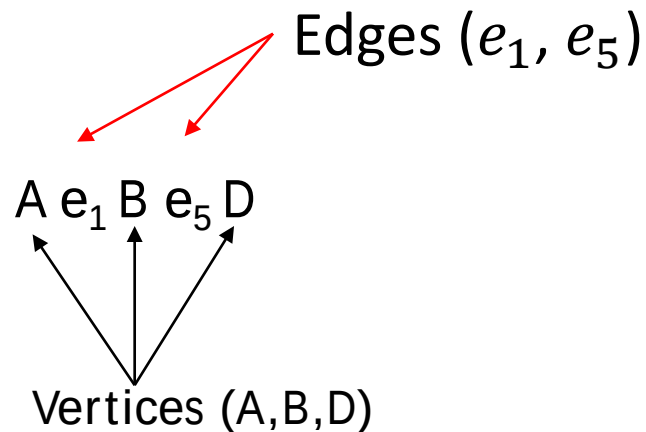
Example (Walk)

(a) Find a walk from A to D.



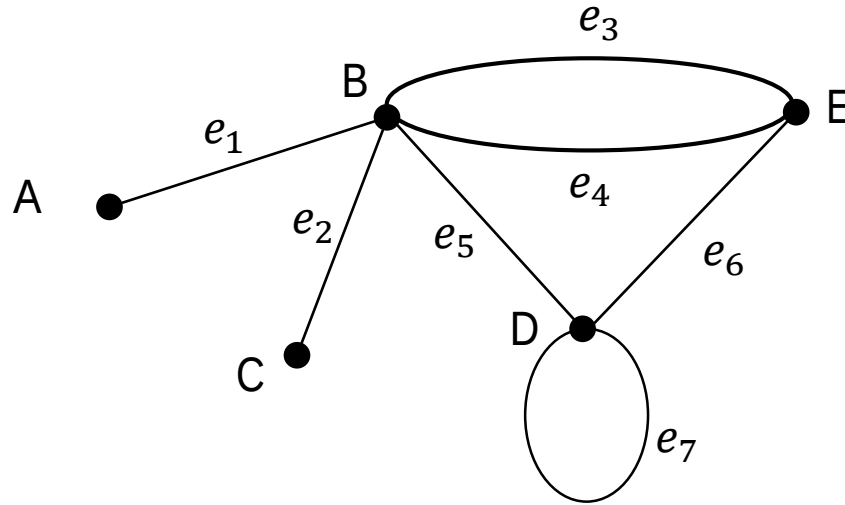
Solution :

We can move from A to B to D, and we write it in this form :



Example (Walk) (Cont'd)

(b) Is this the only walk from A to D?



Solution :

No... in fact there are more :

- $A e_1 B e_3 E e_6 D$
- $A e_1 B e_4 E e_6 D$
- $A e_1 B e_3 E e_4 B e_5 D$
- $A e_1 B e_3 E e_4 B e_3 E e_6 D$ etc ...
- as long as you are able to reach D from A.

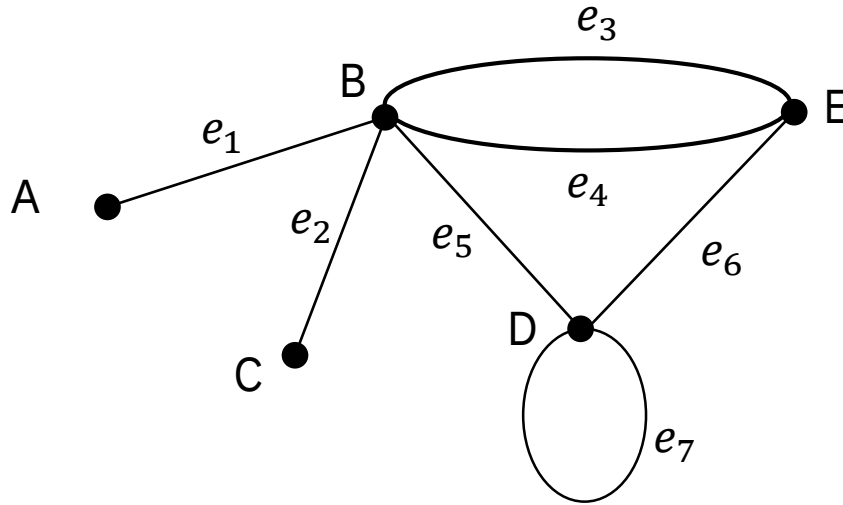
Path

- The second type of graph travel is a path.
- In the previous example, one possible walk that was found was :
 $A e_1 B e_3 E e_4 B e_3 E e_6 D$
- Note that in this walk, there are repeated edges
- What if we want a walk that does not contain any repeated edges ?

A **path** from v to w is a *walk* from v to w that does *not* contain *repeated edges*.

Example (Path)

Find some paths from A to D.



Solution :

- A e_1 B e_3 E e_6 D
- A e_1 B e_4 E e_6 D
- A e_1 B e_3 E e_4 B e_5 D
- A e_1 B e_5 D

No repeated edges

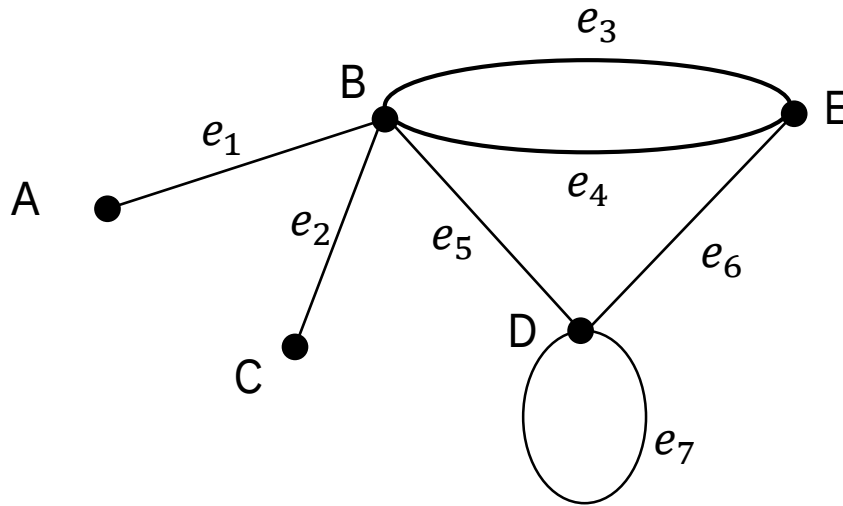
Circuit

- The last type of graph travel is a circuit.
- So far, the two types of graph travel involve moving from one vertex to a *different* vertex.
- What if we want to travel from one vertex and end up back in the *original* vertex that we started from?
- In addition, we *don't* want to *repeat* any edges

A circuit is a *path* that starts and ends at the *same vertex*.

A trivial circuit is a *walk* that has a *single vertex* and *no edge*.

Example (Circuit)



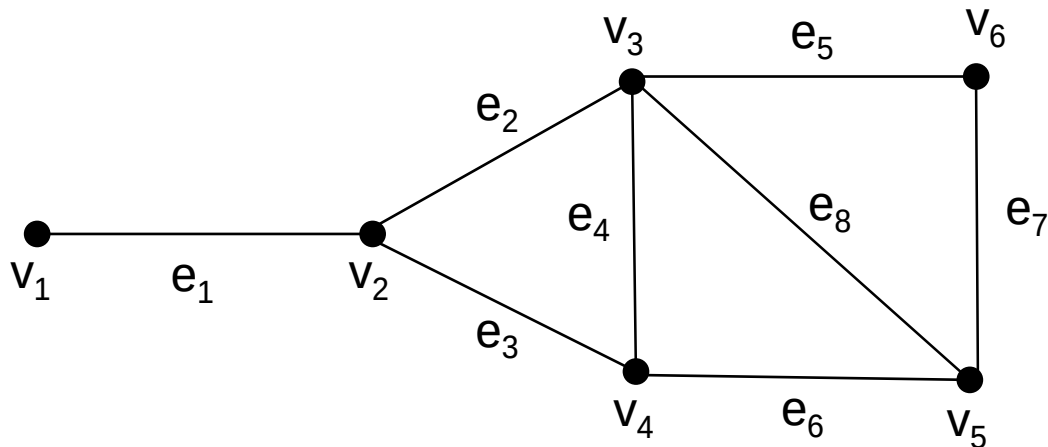
- (a) Is A to D a circuit ?
- (b) Is $A e_1 B e_3 E e_4 B e_1 A$ a circuit?
- (c) Are there any circuit in this diagram ?

Solution :

- (a) No, because it does not start and end at the same vertex.
- (b) No, because e_1 is a repeated edge.
- (c) Yes - $B e_3 E e_4 B$, $D e_7 D$, $D e_5 B e_3 E e_6 D$, $D e_5 B e_4 E e_6 D$

Exercise (Walk, Path, Circuit)

Consider the following graph, determine if the following walks are paths, circuits or just walks. Explain your choice.



	Repeated edges?	Start & end at same point?	Walk / Path / Circuit?
(a) $v_3 e_5 v_6 e_7 v_5 e_6 v_4$	No	No	Path
(b) $v_3 e_5 v_6 e_7 v_5 e_6 v_4 e_4 v_3$	No	Yes	Circuit
(c) $v_3 e_5 v_6 e_7 v_5 e_6 v_4 e_4 v_3 e_8 v_5 e_7 v_6$	Yes	No	Walk

Connected Graphs

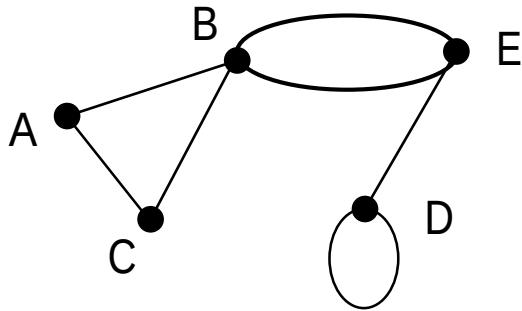
- Intuitively, a graph is connected if you can travel from any vertex to any other vertex along a sequence of adjacent edges.

A graph is connected if, given *any* two vertices v and w , there is a *walk* from v to w .

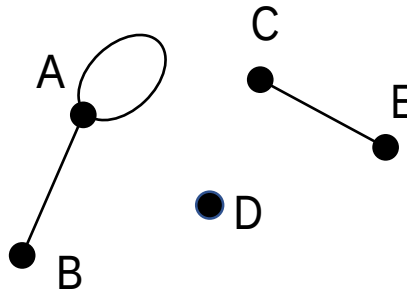
Example (Connected Graphs)

Which of the following are connected graphs? Explain.

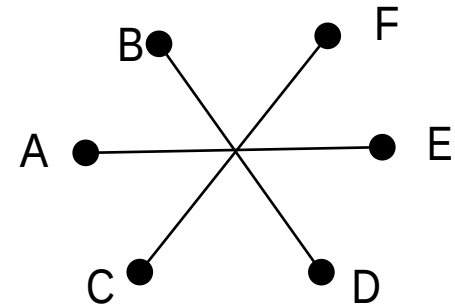
(a)



(b)



(c)

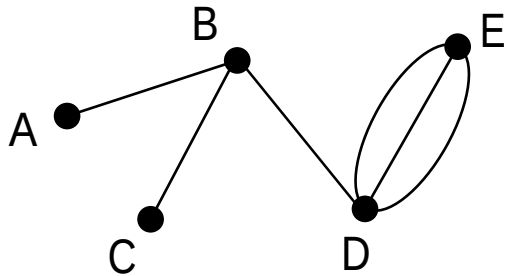


	Connected?	Reason
(a)	Yes	There is a walk between any vertex to any other vertex
(b)	No	There is no walk between A to C,D,E.
(c)	No	There is no walk between A to B,C,D,E,F.

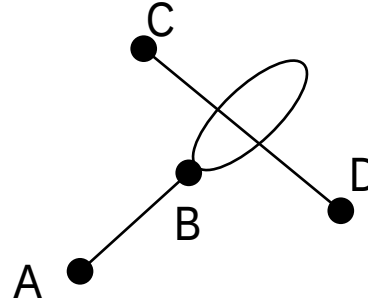
Exercise (Connected Graphs)

Which of the following are connected graphs? Explain.

(a)



(b)



	Connected?	Reason
(a)	Yes	There is a walk between any vertex to any other vertex
(b)	No	There is no walk between A to C,D.

Part 2 Trees

Section 1 Introduction

- Introduction
- Tree vs Non-Tree

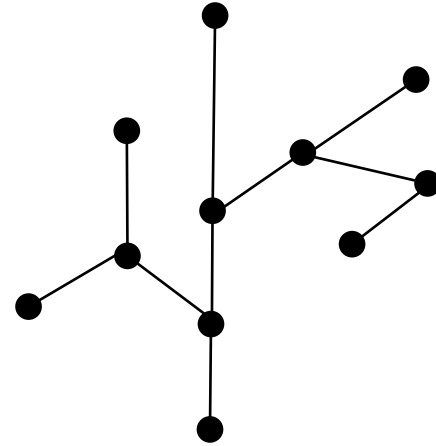
Introduction to Trees

Questions

- Are there any circuit in this graph?
- Is it connected?

Answers

- No circuits exist (Verify it!)
 - Connected (Verify that there is a walk between any two vertices!)
-
- In this section, we will look at this particular type of graph that is circuit-free and is connected.



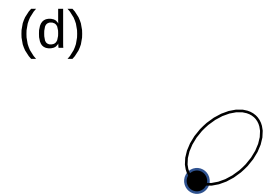
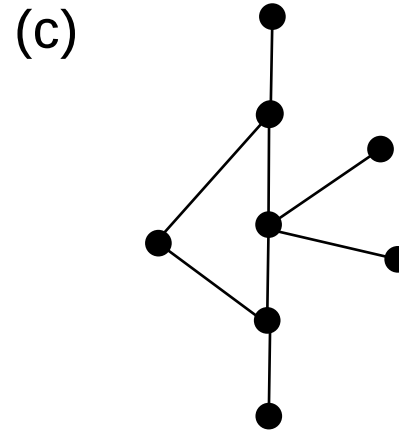
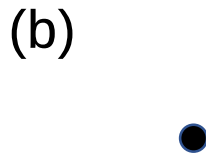
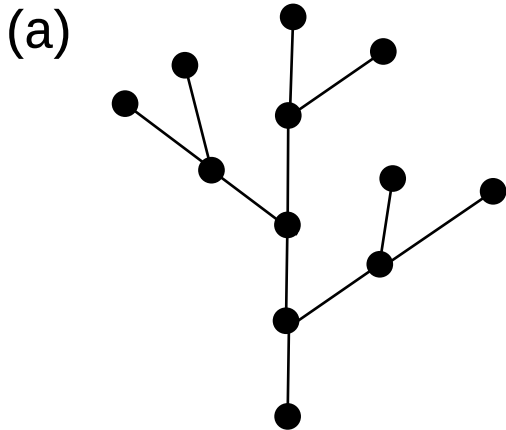
Definition of Trees

A tree is a graph that has no non-trivial circuits and is connected.

A trivial tree is a graph that is has a single vertex.

Examples (Tree vs Non-Tree)

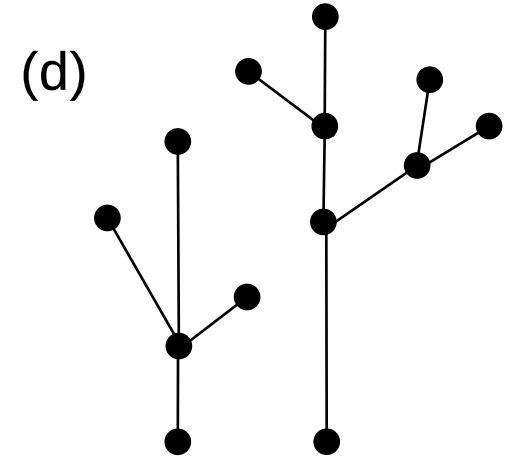
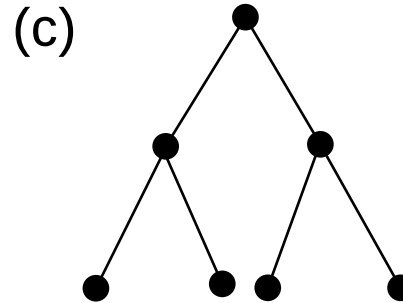
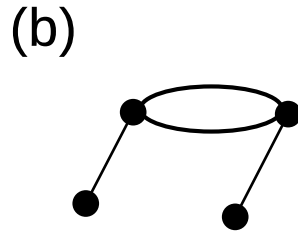
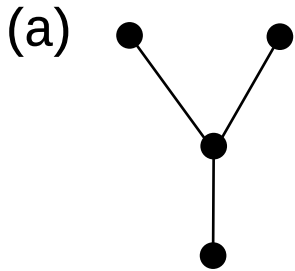
Which of the following are trees? Explain.



	Tree?	Reason
(a)	Yes	Circuit-free and connected
(b)	Yes	Circuit-free and connected to itself (trivial tree)
(c)	No	Not circuit-free
(d)	No	Not circuit-free

Exercise (Tree vs Non-Tree)

Which of the following are trees? Explain.



	Tree?	Reason
(a)	Yes	Circuit-free and connected
(b)	No	Not circuit-free
(c)	Yes	Circuit-free and connected
(d)	No	Not connected

Section 2 Rooted Trees

- Introduction to Rooted Trees
- Terminologies in Rooted Trees
- Decision Trees
- Parse Trees
- Binary Trees

Introduction to Rooted Trees

We will now focus on a particular type of tree called the rooted tree.

We start off by defining :

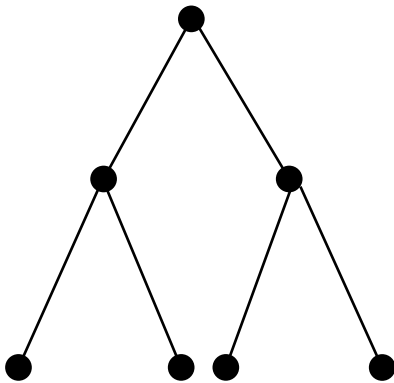
- Terminologies of Rooted Trees

This will be followed by some special rooted trees and their applications.

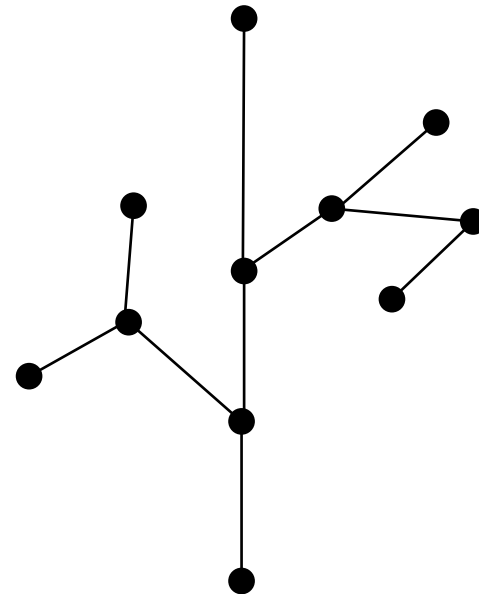
- Decision Trees
- Parse Trees
- Binary Trees

Definition of Rooted Trees

A rooted tree is a tree in which there is one vertex (root) that is distinguished from the others.



Rooted Tree

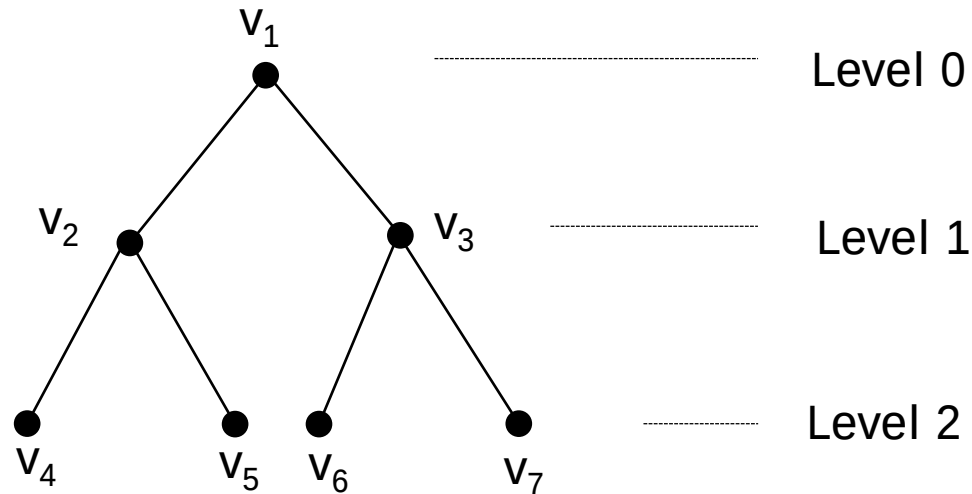


Non-Rooted Tree

Terminologies in Rooted Trees

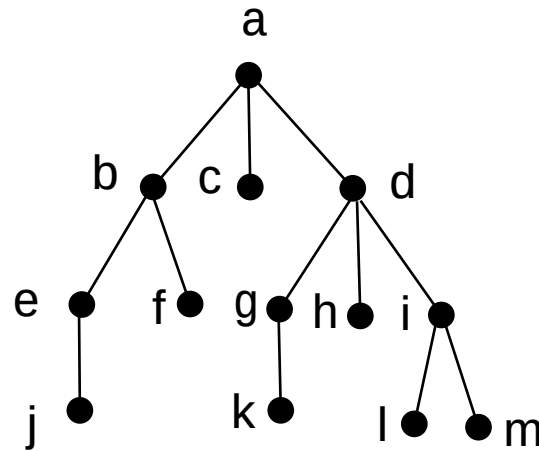
- The level of a vertex
 - is the number of edges along the unique path between it and the root.
- The height of a rooted tree
 - is the maximum level of any vertex of a tree.
- A vertex is a child of another vertex
 - if they are connected by an edge and is 1 level below.
- A vertex is a parent of another vertex
 - if they are connected by an edge and is 1 level above.
- A vertex is a sibling of another vertex
 - if the two vertices are children of the same parent.

Example (Terminologies in Rooted Trees)



- v_1 is the root.
- The height of the tree is 2.
- v_5 is in level 2.
- v_1 is the parent of v_2 and v_3 .
- v_2 and v_3 are the child of v_1 .
- v_7 is the sibling of v_6 .
- $v_2, v_3, v_4, v_5, v_6, v_7$ are the descendants of v_1 .

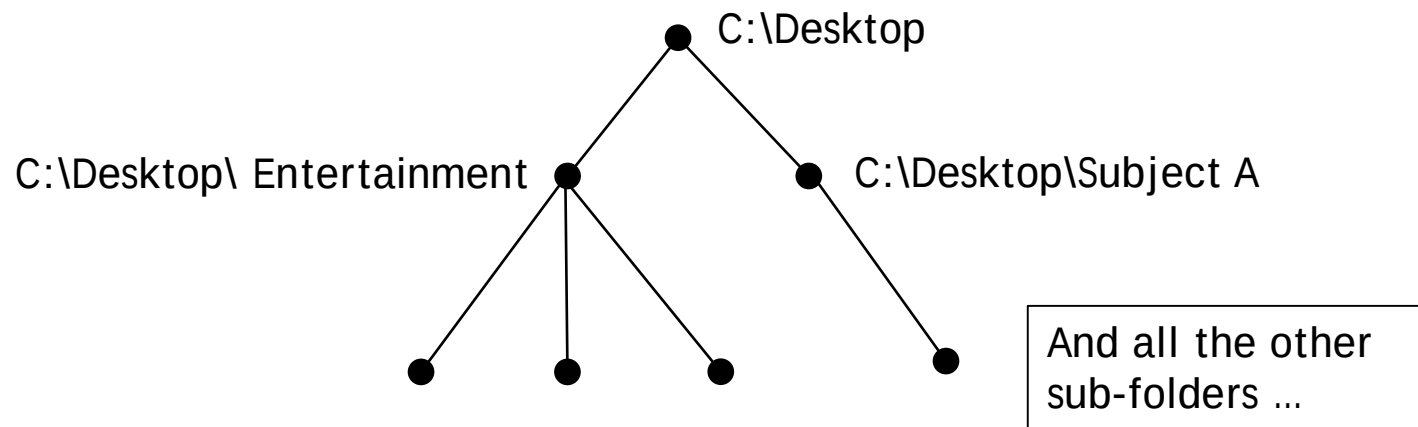
Exercise (Terminologies in Rooted Trees)



(a)	What is the level of d?	1
(b)	What is the level of m?	3
(c)	What is the height of the rooted tree?	3
(d)	What are the children of i?	l and m
(e)	What is the parent of f?	b
(f)	What is the siblings of h?	g and i
(g)	What are the descendants of b?	e, f, j

Application : Rooted Tree in File System

The file system in our operating systems can be modelled as a rooted tree structure.



Decision Trees

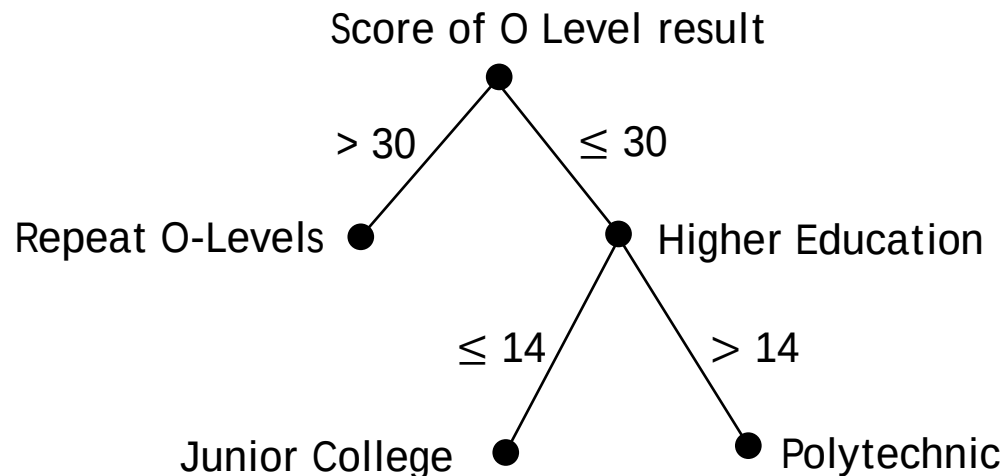
A Decision Tree is a rooted tree that

- lists out all possible consequences or events that can happen in a scenario.
- is used extensively in computer algorithms (such as Classification in data mining) and artificial intelligence.

Application :

Decision Tree in Classification of Data Mining

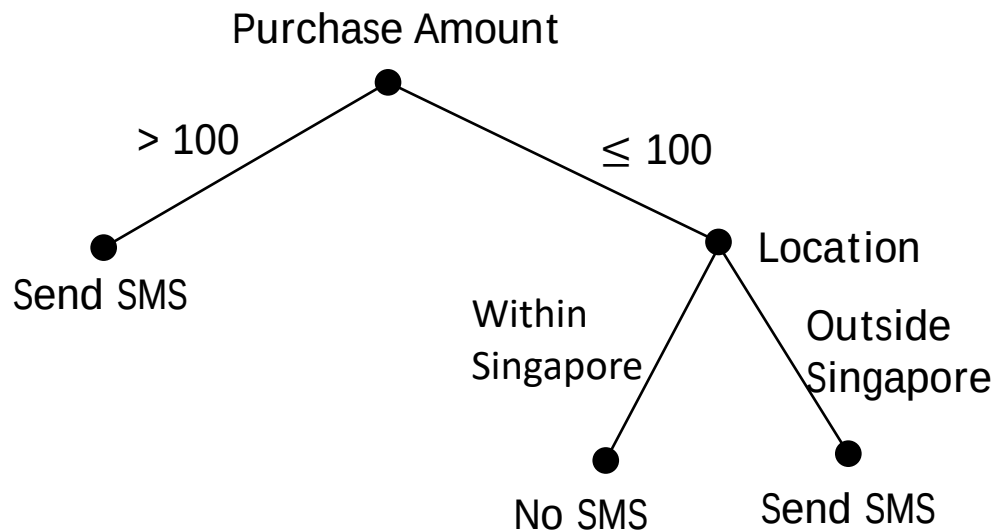
The following algorithm is used to make a recommendation to a student waiting for his O-level results :



(a)	What path should a student with 17 points take?	Polytechnic
(b)	What path should a student with 32 points take?	Repeat O-Levels

Exercise (Applying Decision Tree to Classification in Data Mining)

An SMS is sent to the credit card holder if the purchase amount is more than \$100 or if the purchase is made outside of Singapore. Draw a decision tree to help you write the algorithm.



(a)	What if the purchase is \$65?	No SMS
(b)	What if the purchase is \$5 and location is Null Island** ?	Send SMS

** Where in the world is Null Island? Go find out!

Parse Trees

A Parse Tree is a rooted tree that

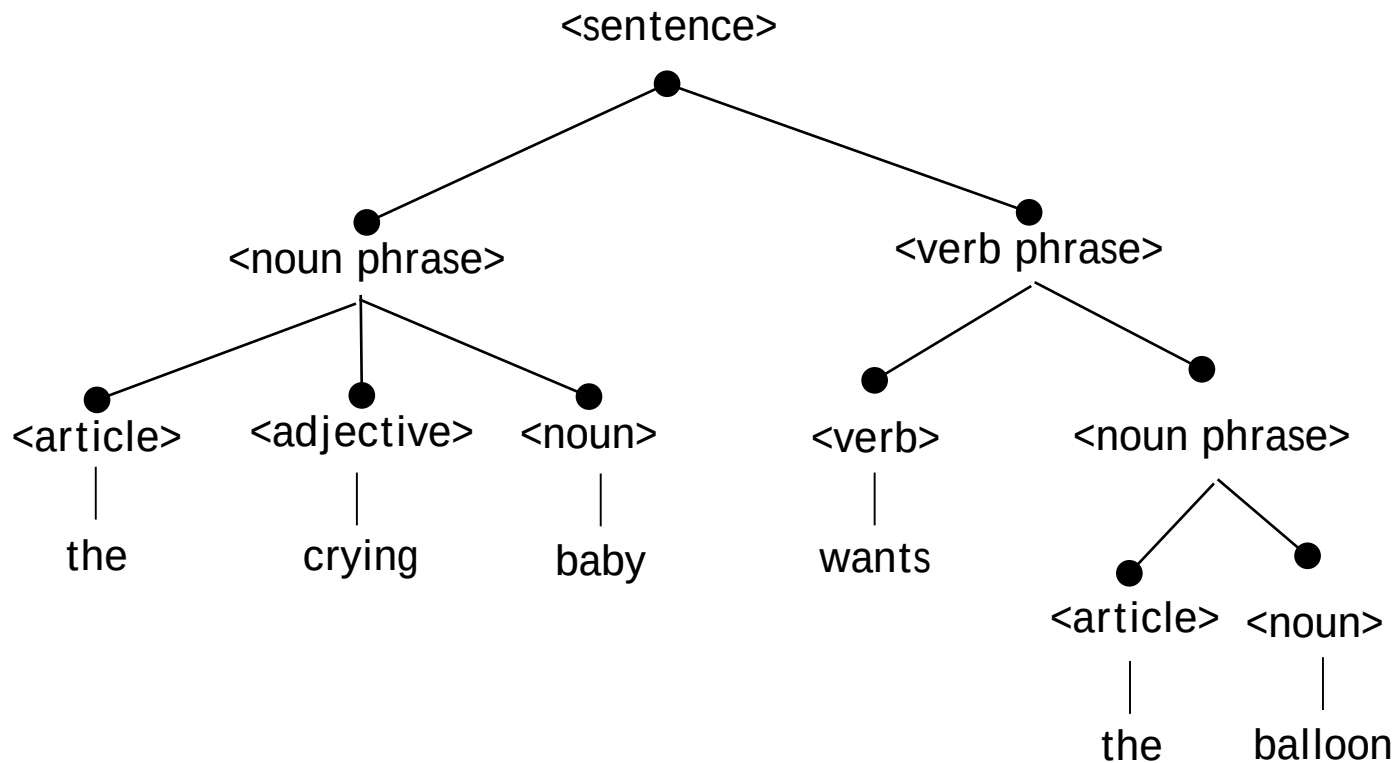
- Shows how grammatically correct sentences (syntax) are derived based on certain rules.
- is used extensively in Natural Language Processing and in Text Mining.
- A subset of the English grammar has the following rules :

A <sentence> is made up of a <noun phrase> followed by a <verb phrase>;

- a. A <noun phrase> is made up of
 - i. an <article> followed by a <noun>; or
 - ii. an <article> followed by an <adjective> followed by a <noun>;
- b. A <verb phrase> is made up of a <verb> followed by a <noun phrase>.

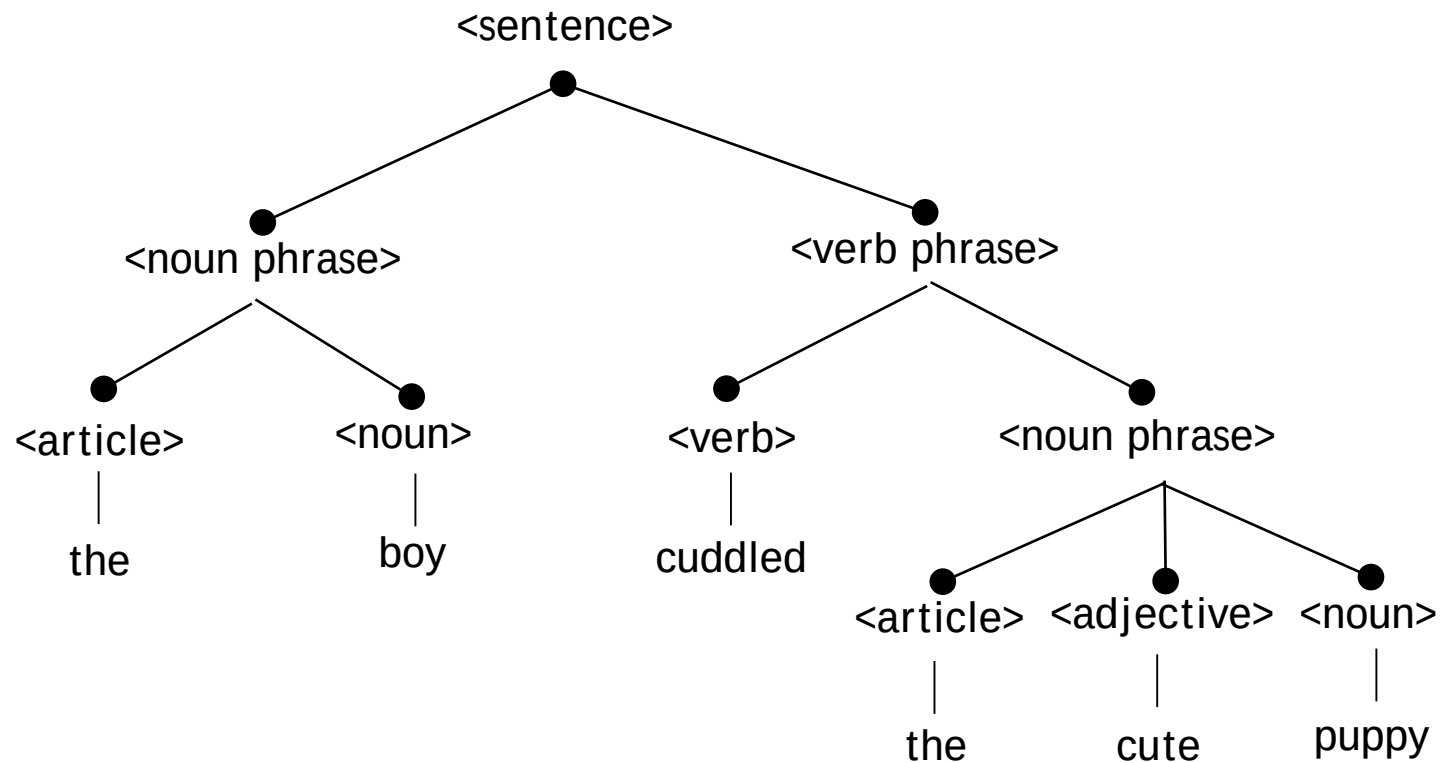
Application : Parse Trees in Text Mining

Draw a parse tree to describe the parsing of the sentence :
“The crying baby wants the balloon.”



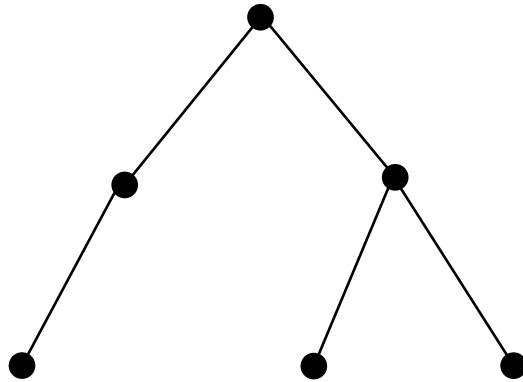
Exercise (Applying Parse Trees to Text Mining)

Draw a parse tree to describe the parsing of the sentence :
“The boy cuddled the cute puppy.”



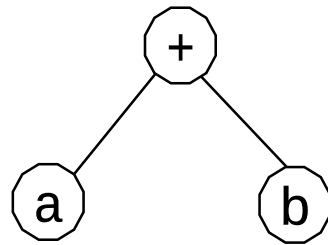
Binary Trees

- A special kind of rooted tree is known as the binary tree.
- This is used extensively in data structures in computing science.
- A binary tree is a rooted tree in which every parent has at most 2 children.

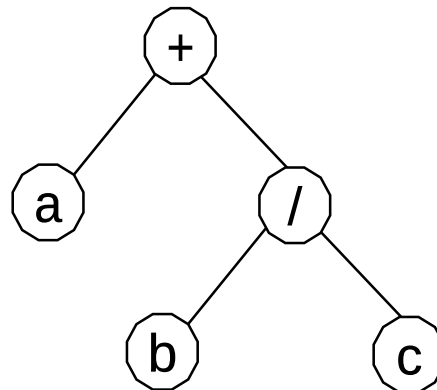


Application : Binary Trees in Algebraic Expressions

- Binary trees are used in computer science to represent algebraic expressions.
- The expression $a + b$ is represented as follows :



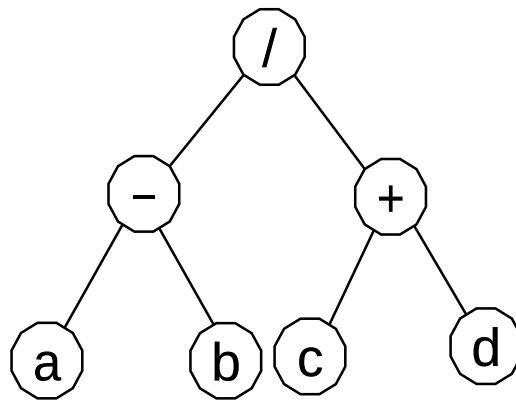
- The expression $a + (b/c)$ is represented as follows :



Exercise (Apply Binary Trees to Algebraic Expressions)

Draw a binary tree to represent the expressions $(a - b) / (c + d)$.

Solution :



Section 3

Spanning Trees

- Introduction
- Spanning Tree
- Weighted Graph
- Minimum Spanning Tree

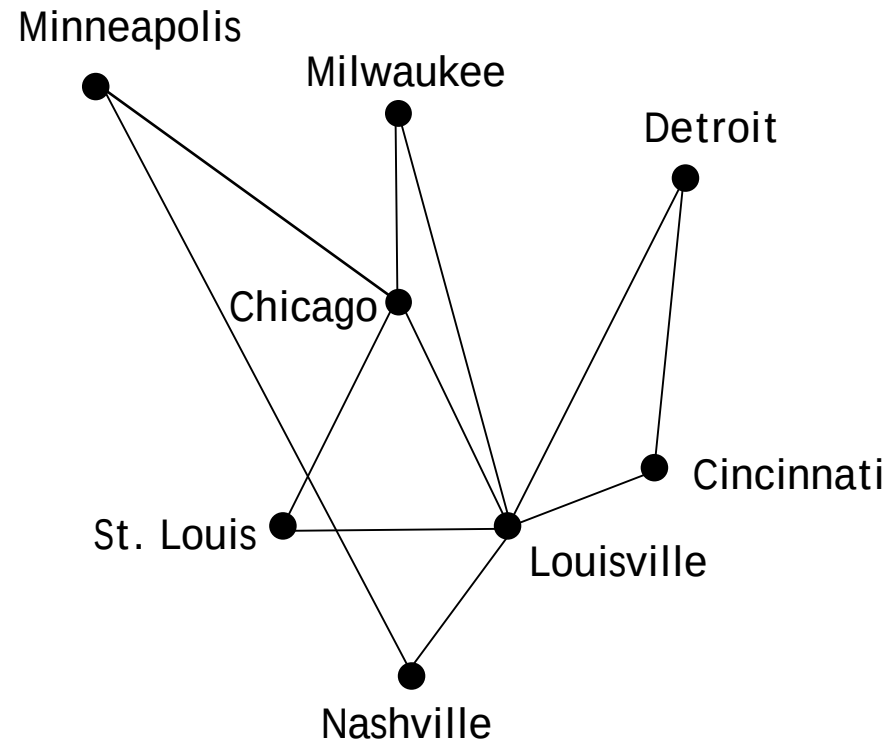
Introduction

Consider the graph on the right.

Suppose an airline wants to :

1. serve all the cities,
2. and, at the same time, use the fewest possible routes to connect them.

How should they go about doing it ?



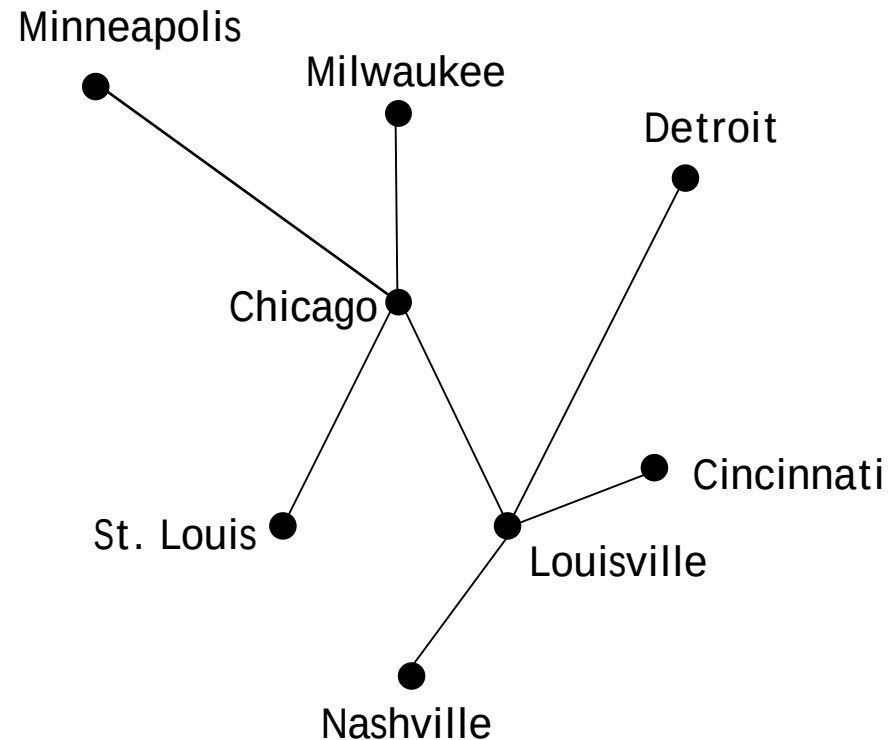
Spanning Tree

To do that, the airline must ensure :

- All vertices (cities) are included.
- All vertices are connected (directly or indirectly) - to travel from any one city to any others.
- There are no circuits – If there is a circuit, then removing one of the edges will reduce the routes, yet will not disconnect the cities.

Such a graph is really just a tree that spans all vertices.

One example is shown on the right.



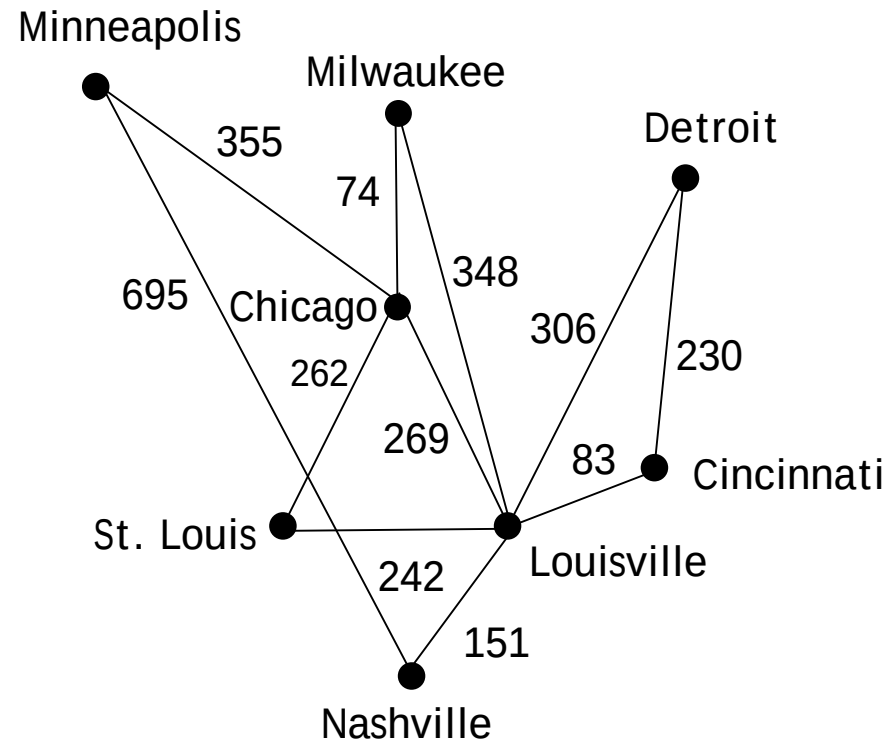
A spanning tree is a tree that contains all vertices of the graph.

Weighted Graph

Suppose that the distance (in miles) is added between each pair of cities.

The number on each edge (distance) is called the **weight**.

The graph becomes a weighted graph.



A **weighted graph** is a graph with edges labelled with numbers (known as weights).

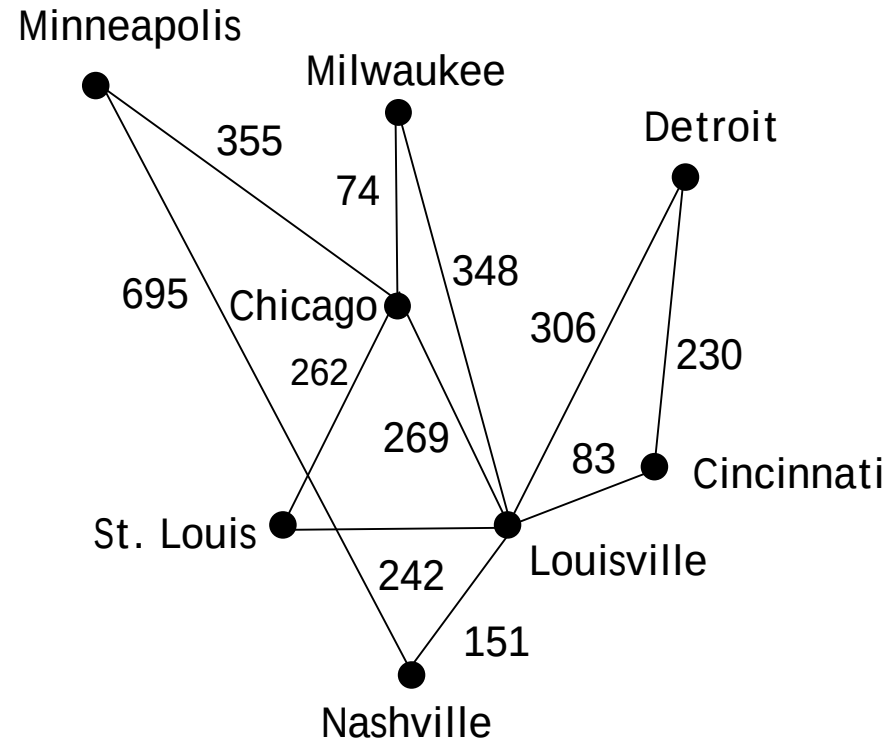
Minimum Spanning Tree

Now, what will happen if the airline wants to simultaneously :

- connect one city to another, and
- minimize the total mileage to save cost and fuel?

As seen earlier, a spanning tree is needed. But this time, the sum of the weights of all the edges must be as small as possible.

This is known as a minimum spanning tree.



A minimum spanning tree is a spanning tree that has the least total weight compared to other spanning trees.

Algorithm To Find Minimum Spanning Tree

An algorithm is a step-by-step set of operations to be performed.

To find the minimum spanning tree, we will use [Kruskal's algorithm](#) where edges of the weighted graph are considered one by one in order of increasing weight.

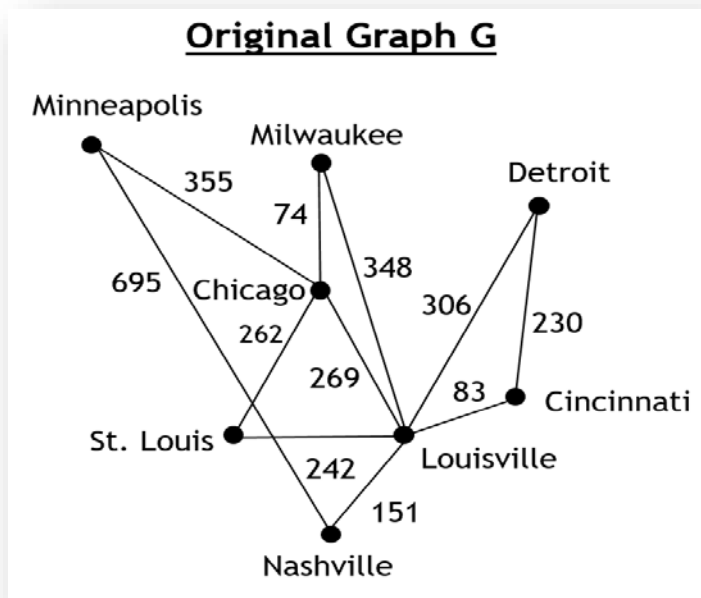
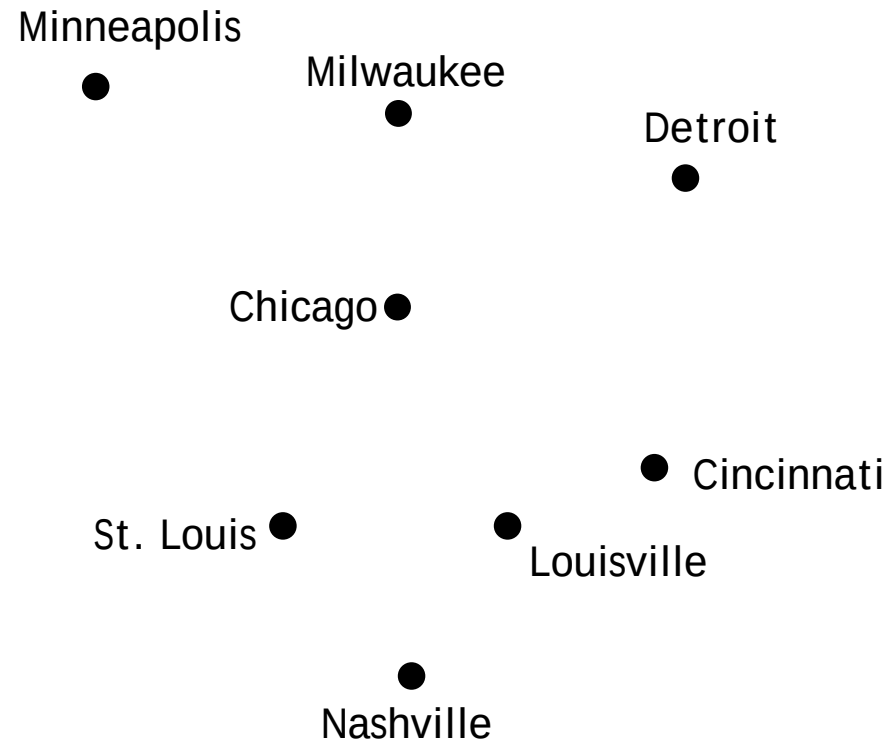
1. Create a new graph.
Initialize it with only the vertices from the original graph G .
2. Let $E(G)$ be the set of all the edges of G
Let $m = 0$ and let $n = \text{number of vertices}$
3. While ($m < n - 1$)
 - a) Find an edge e in set $E(G)$ with the least weight.
 - b) Delete e from set $E(G)$.
 - c) If the addition of e to the new graph does not produce a circuit, then add e to the graph and set $m = m + 1$.

Application : Minimum Spanning Tree in Communication System

Find the minimal spanning tree for the airline example.

Step 1

Initialize the graph to contain only the vertices in the original graph G.

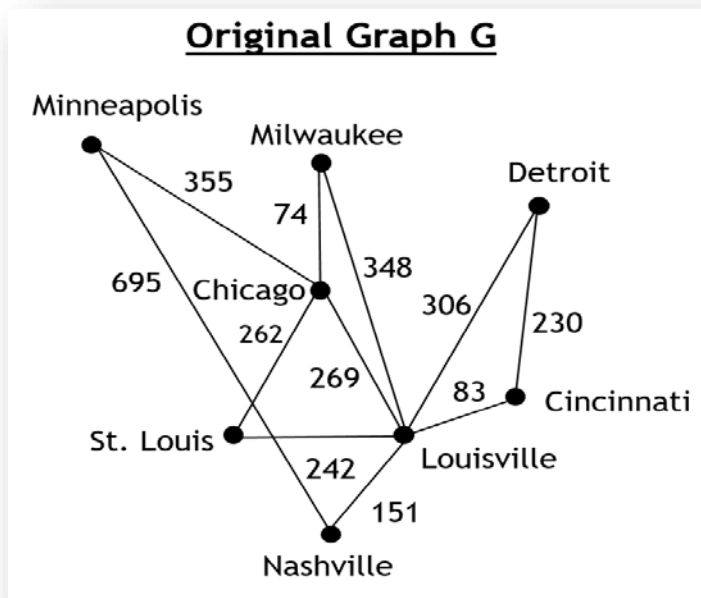
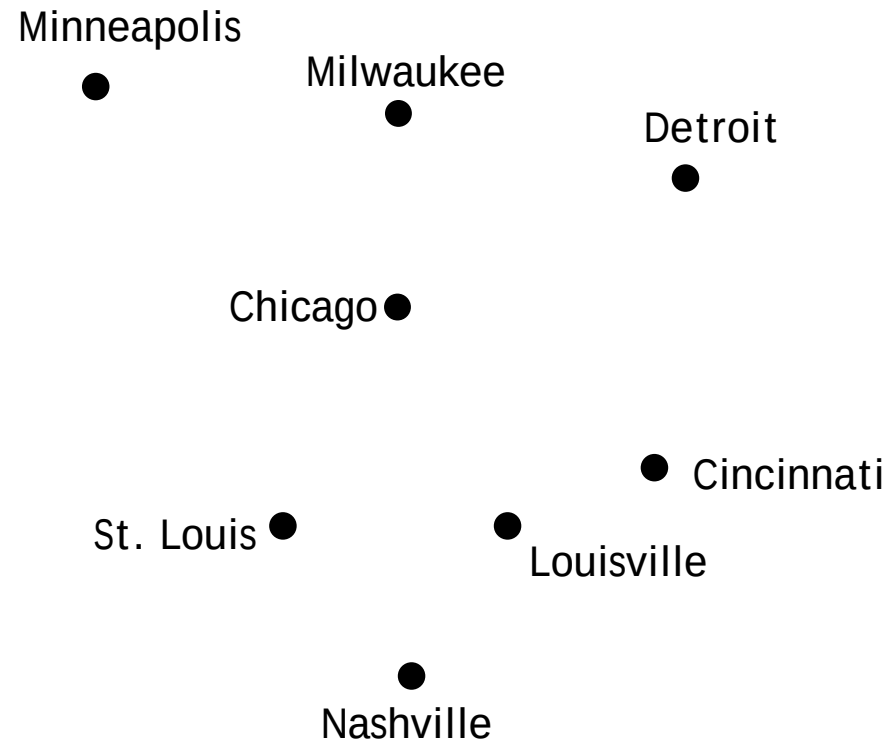


Application : Minimum Spanning Tree in Communication System (Cont'd)

Find the minimal spanning tree for the airline example.

Step 2

Let $E(G)$ be the set of all edges of G .
Let $m = 0$ and $n = \text{number of vertices}$.
In this case, $m = 0$ and $n = 8$.



Application : Minimum Spanning Tree in Communication System (Cont'd)

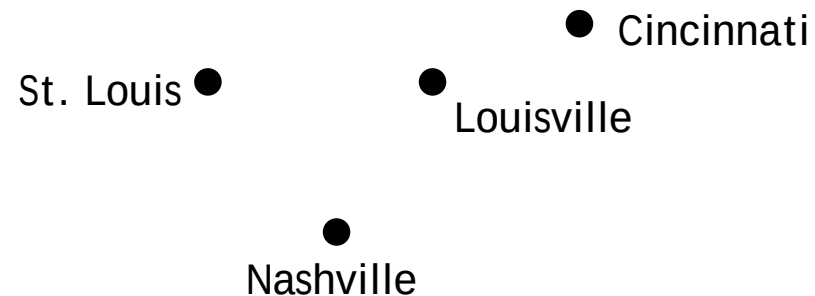
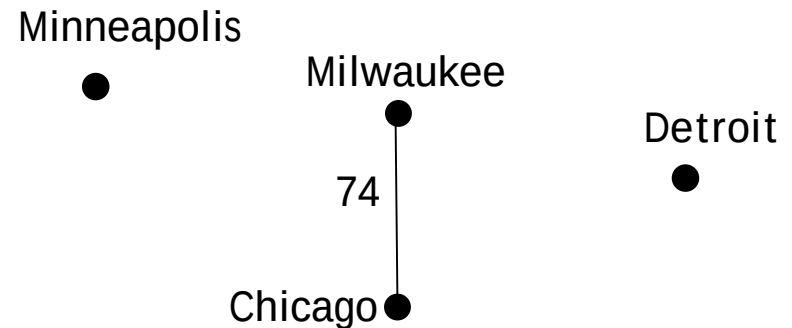
Find the minimal spanning tree for the airline example.

Step 3

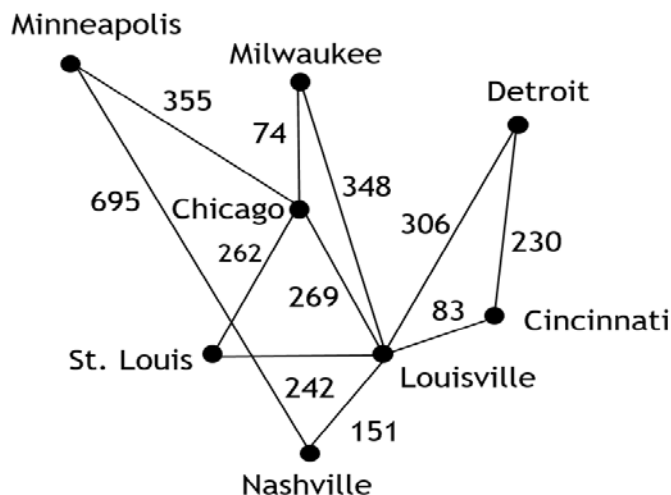
Locate the edge with the least weight :

Chicago to Milwaukee (74)

Add the edge and update to $m = 1$.



Original Graph G



Application : Minimum Spanning Tree in Communication System (Cont'd)

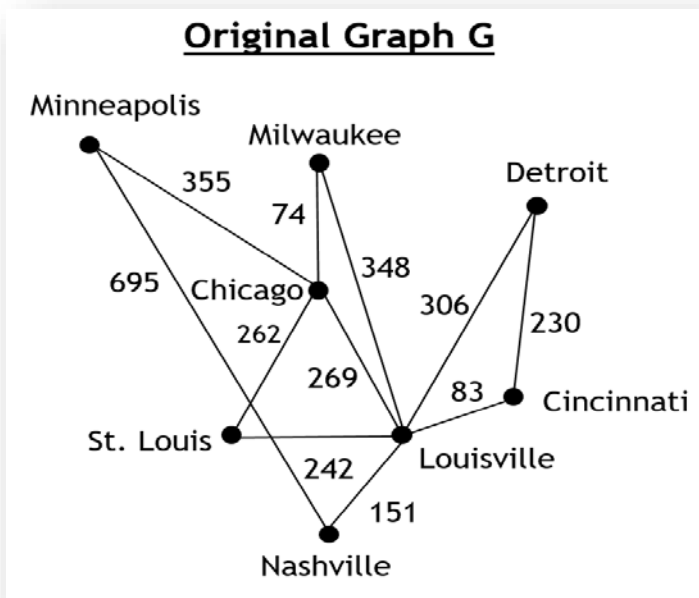
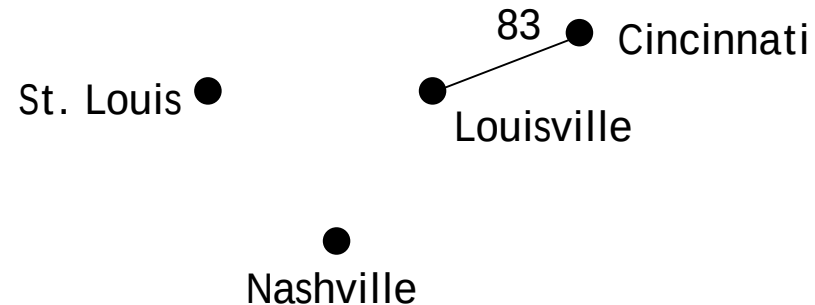
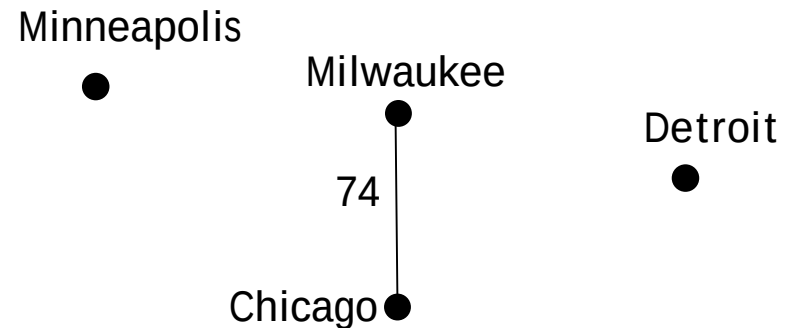
Find the minimal spanning tree for the airline example.

Step 4

Locate the next edge with the least weight :

Louisville to Cincinnati (83)

Add the edge and update to $m = 2$.



Application : Minimum Spanning Tree in Communication System (Cont'd)

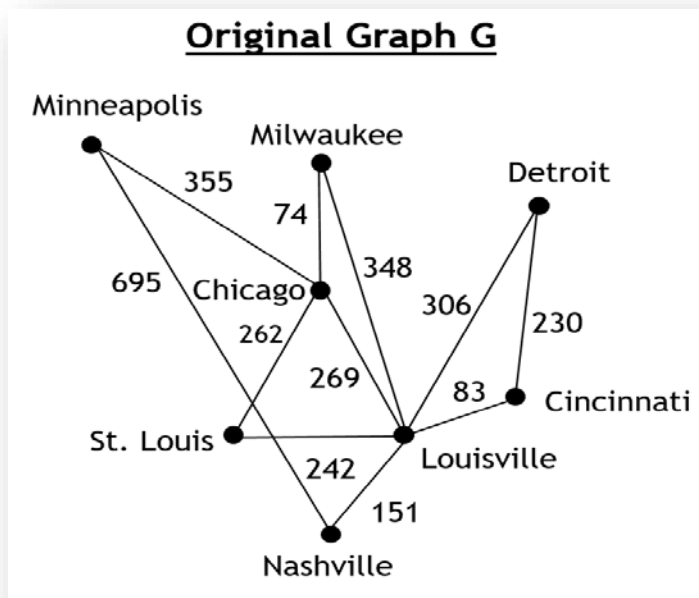
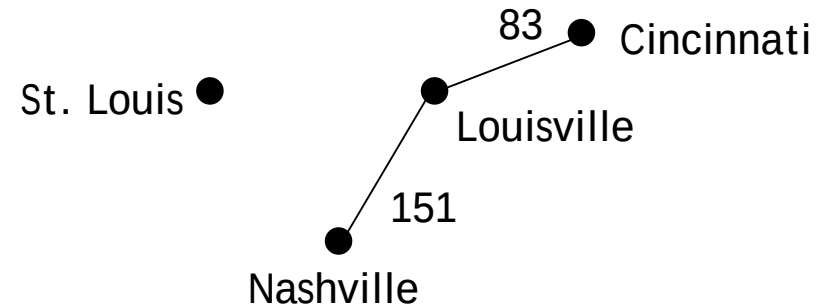
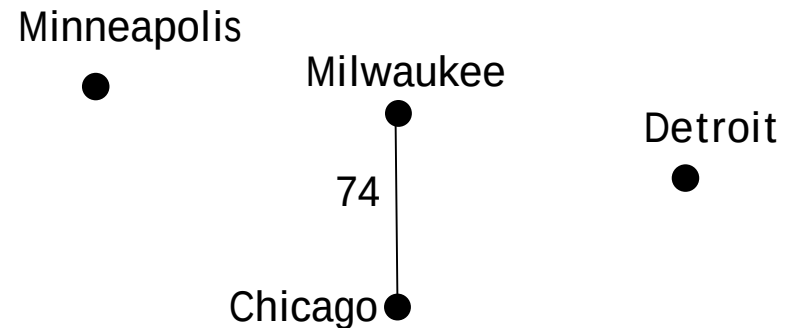
Find the minimal spanning tree for the airline example.

Step 5

Locate the next edge with the least weight :

Louisville to Nashville (151)

Add the edge and update to $m = 3$.



Application : Minimum Spanning Tree in Communication System (Cont'd)

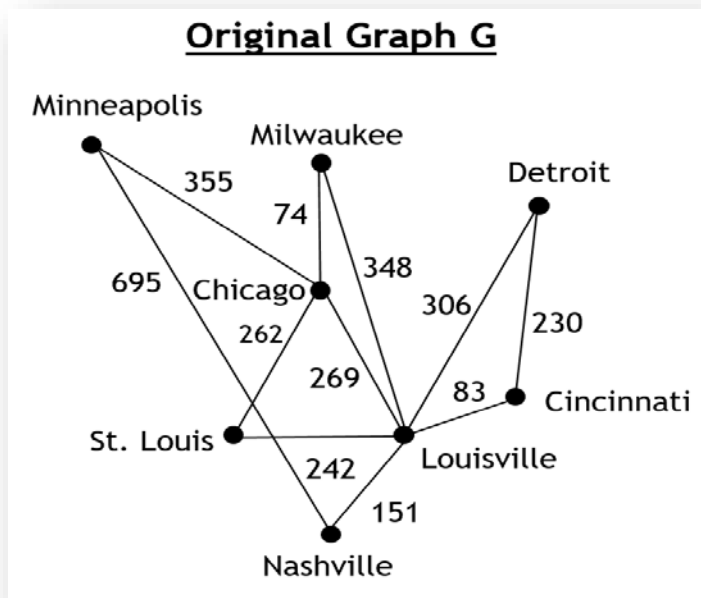
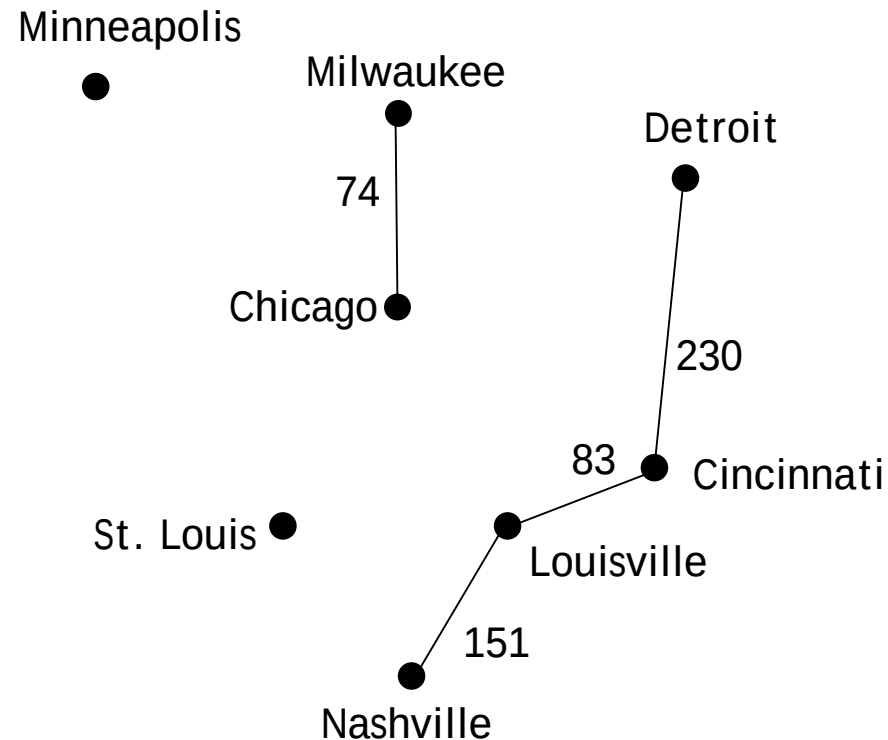
Find the minimal spanning tree for the airline example.

Step 6

Locate the next edge with the least weight :

Cincinnati to Detroit (230)

Add the edge and update $m = 4$.



Application : Minimum Spanning Tree in Communication System (Cont'd)

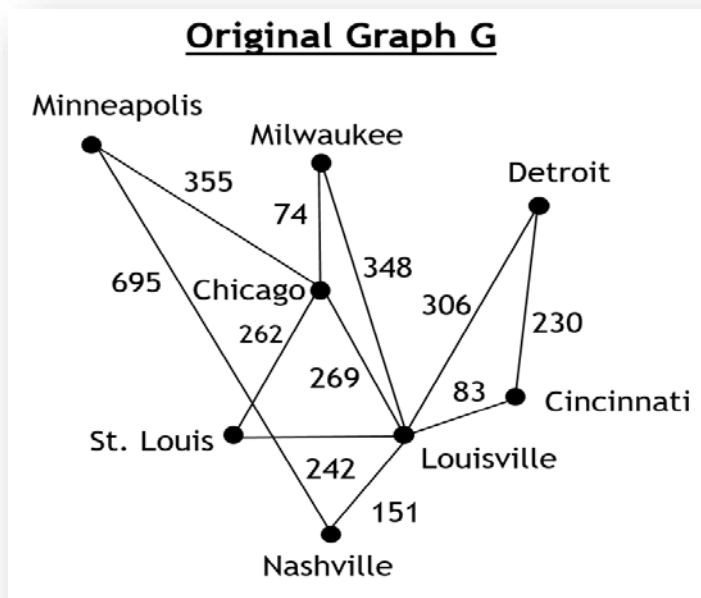
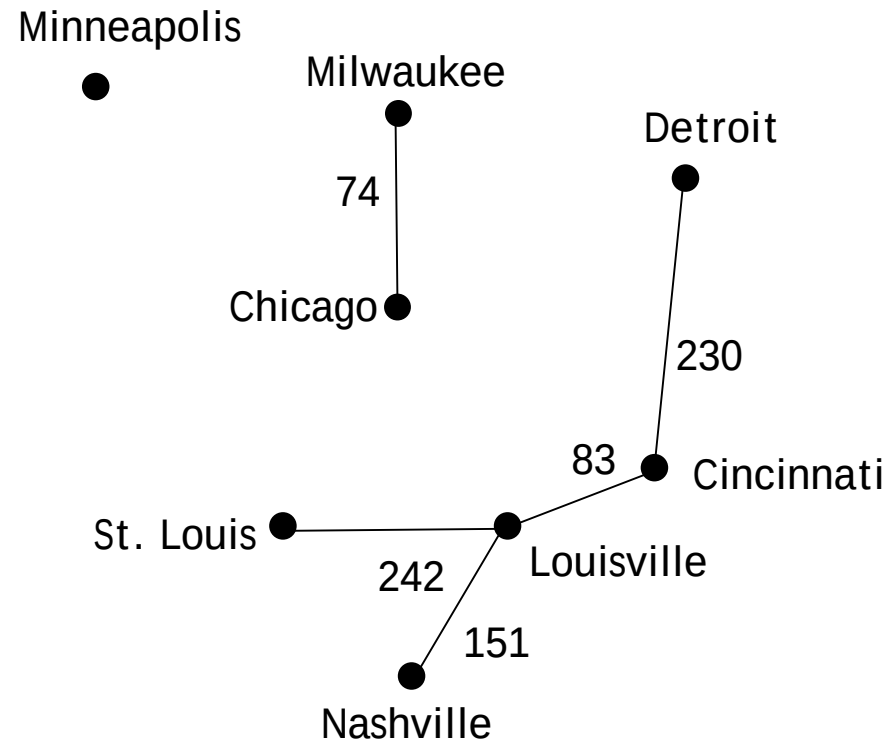
Find the minimal spanning tree for the airline example.

Step 7

Locate the next edge with the least weight

Louisville to St. Louis (242)

Add the edge and update to $m = 5$.



Application : Minimum Spanning Tree in Communication System (Cont'd)

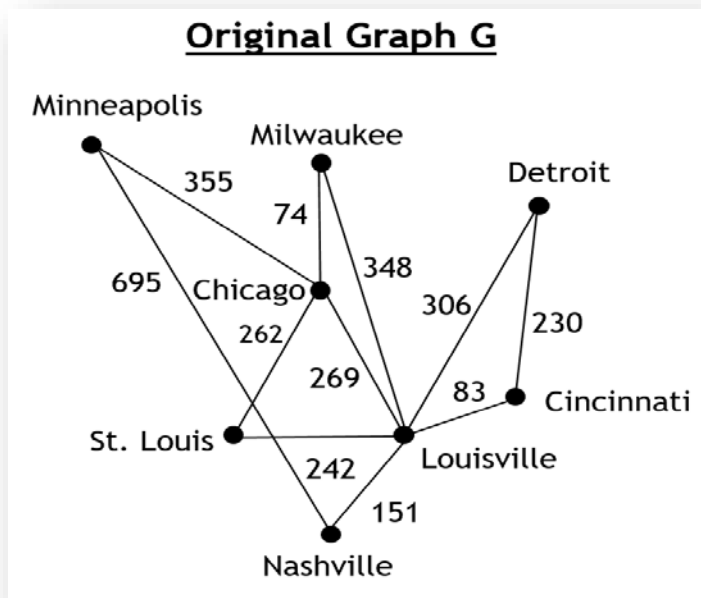
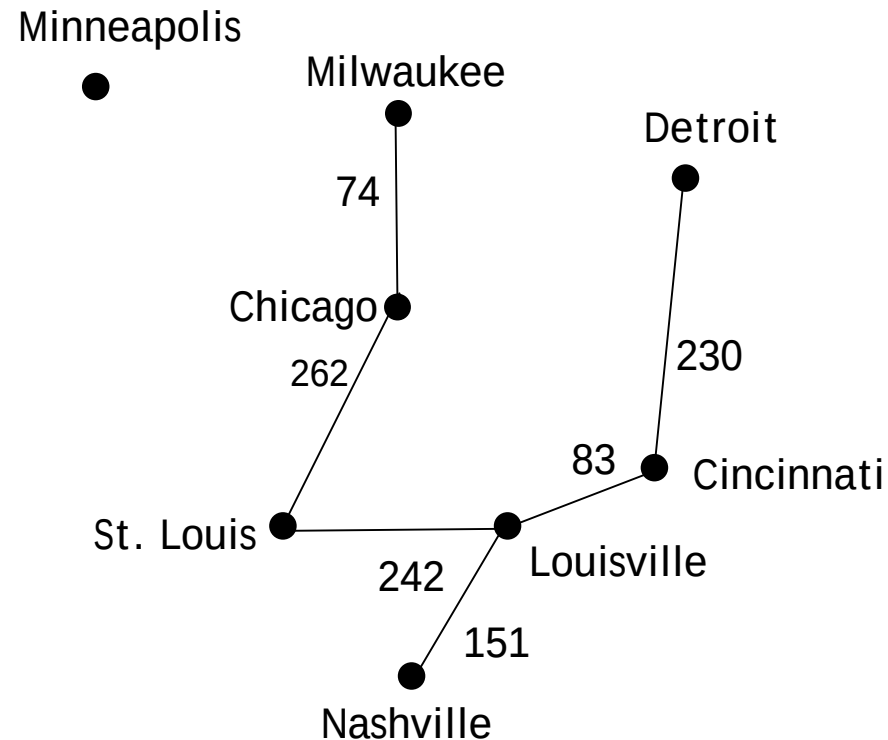
Find the minimal spanning tree for the airline example.

Step 8

Locate the next edge with the least weight :

Chicago to St. Louis (262)

Add the edge and update to $m = 6$.



Application : Minimum Spanning Tree in Communication System (Cont'd)

Find the minimal spanning tree for the airline example.

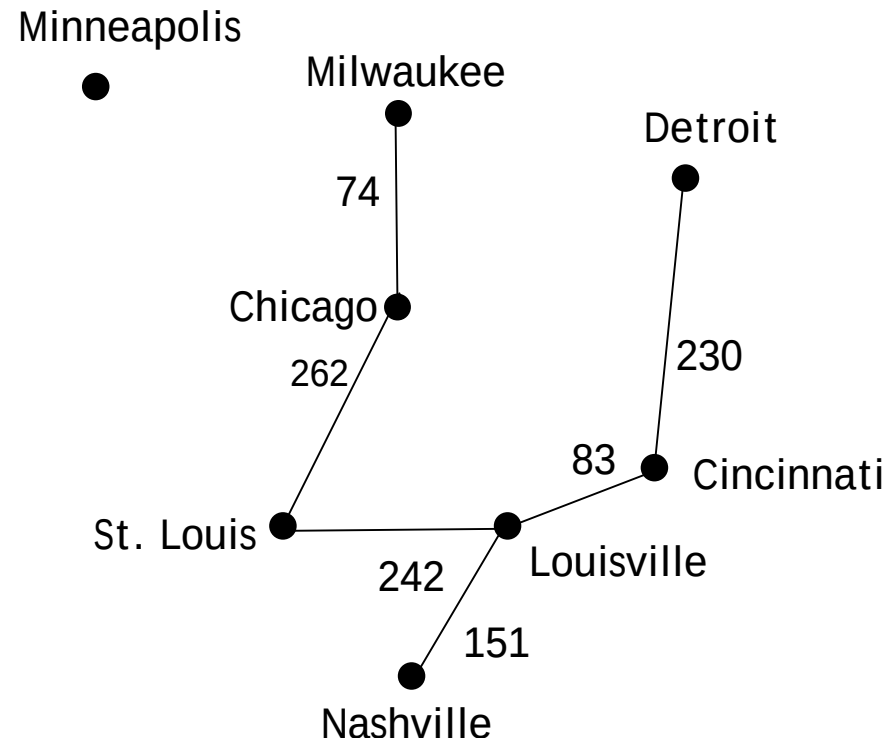
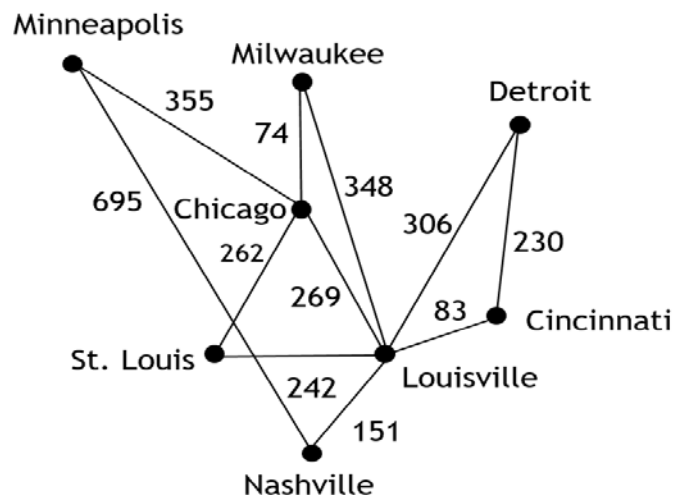
Step 9

Locate the next edge with the least weight :

Chicago to Louisville (269)

But adding this edge will result in a circuit. Therefore, we move on.

Original Graph G



Application : Minimum Spanning Tree in Communication System (Cont'd)

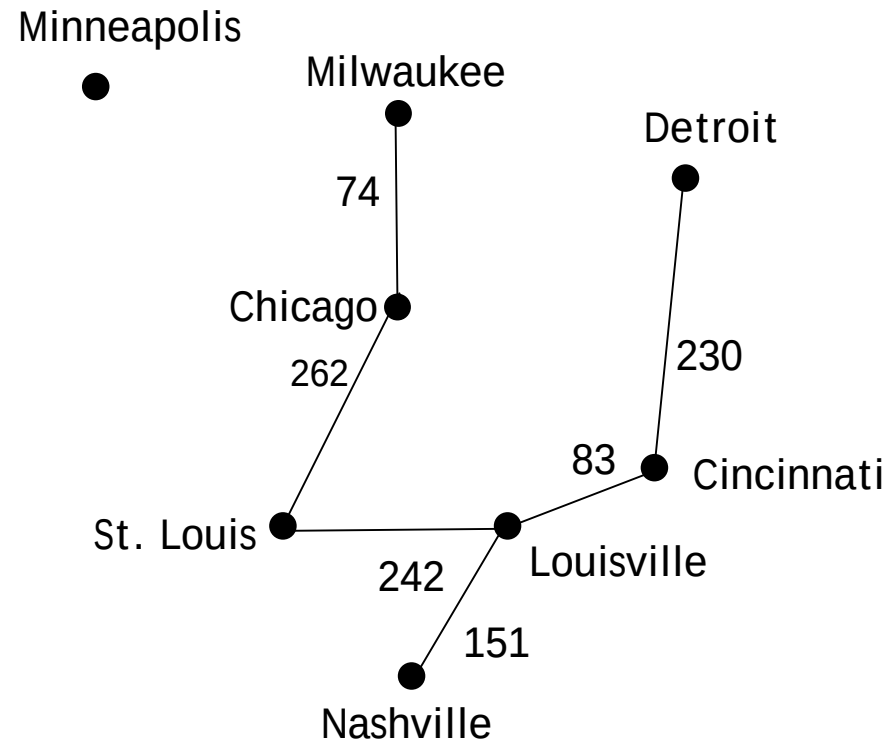
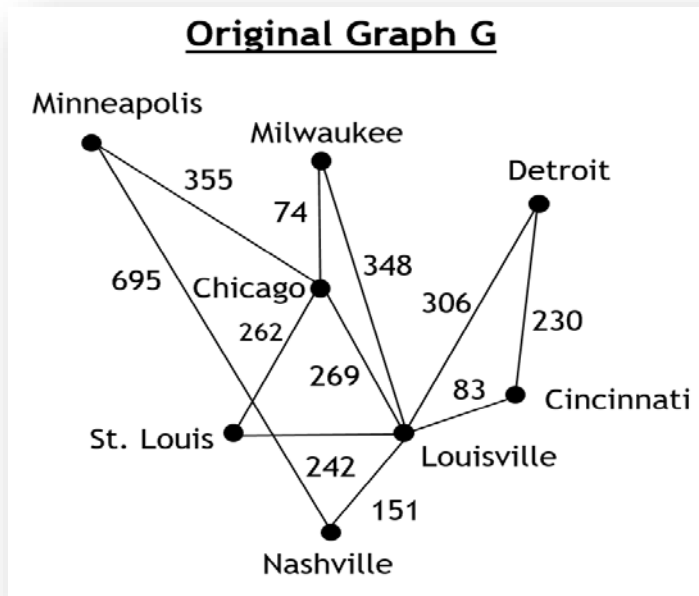
Find the minimal spanning tree for the airline example.

Step 10

Locate the next edge with the least weight :

Detroit to Louisville (306)

But adding this edge will result in a circuit. Therefore, we move on.



Application : Minimum Spanning Tree in Communication System (Cont'd)

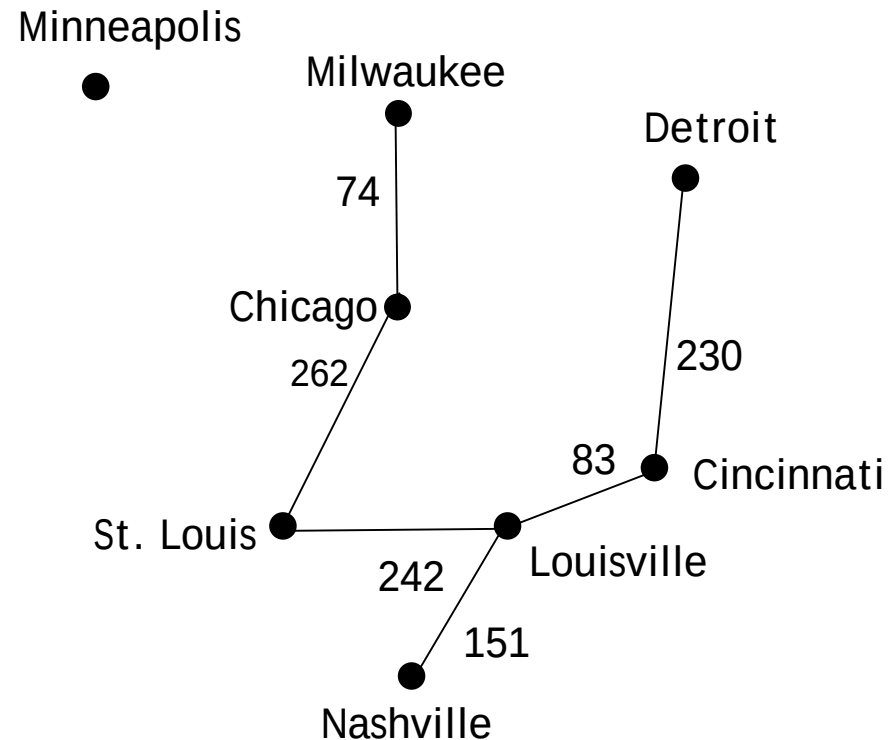
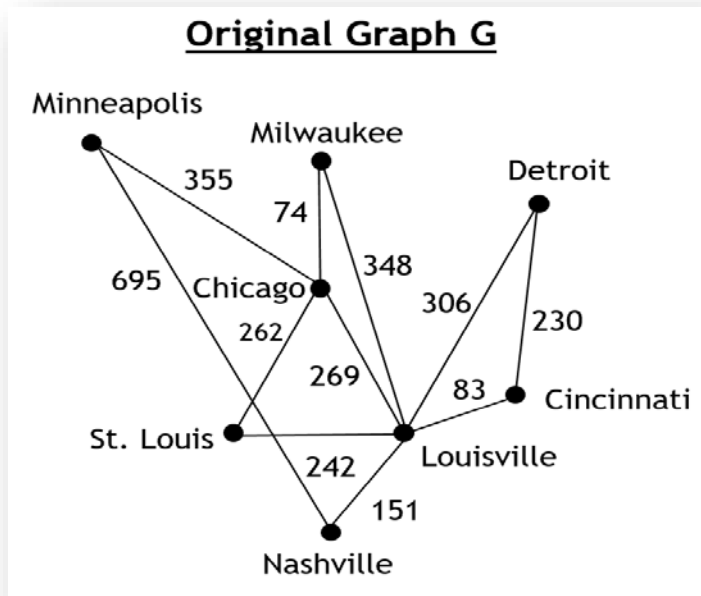
Find the minimal spanning tree for the airline example.

Step 11

Locate the next edge with the least weight :

Louisville to Milwaukee (348)

But adding this edge will result in a circuit. Therefore, we move on.



Application : Minimum Spanning Tree in Communication System (Cont'd)

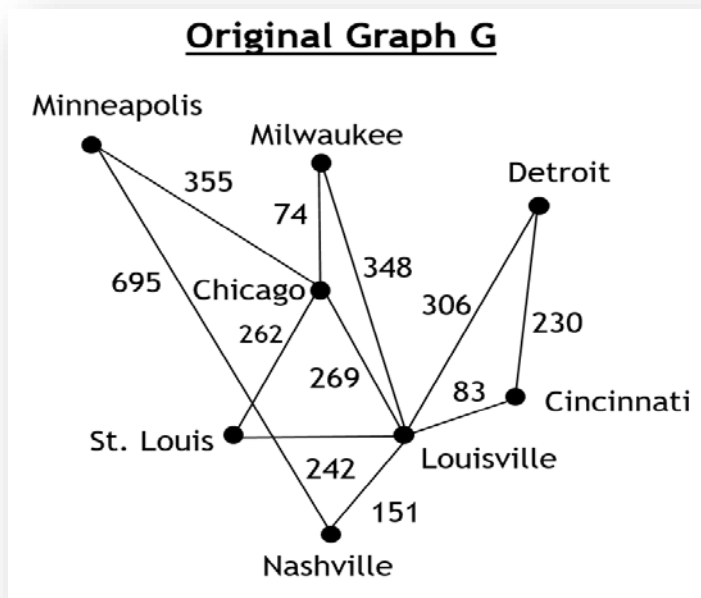
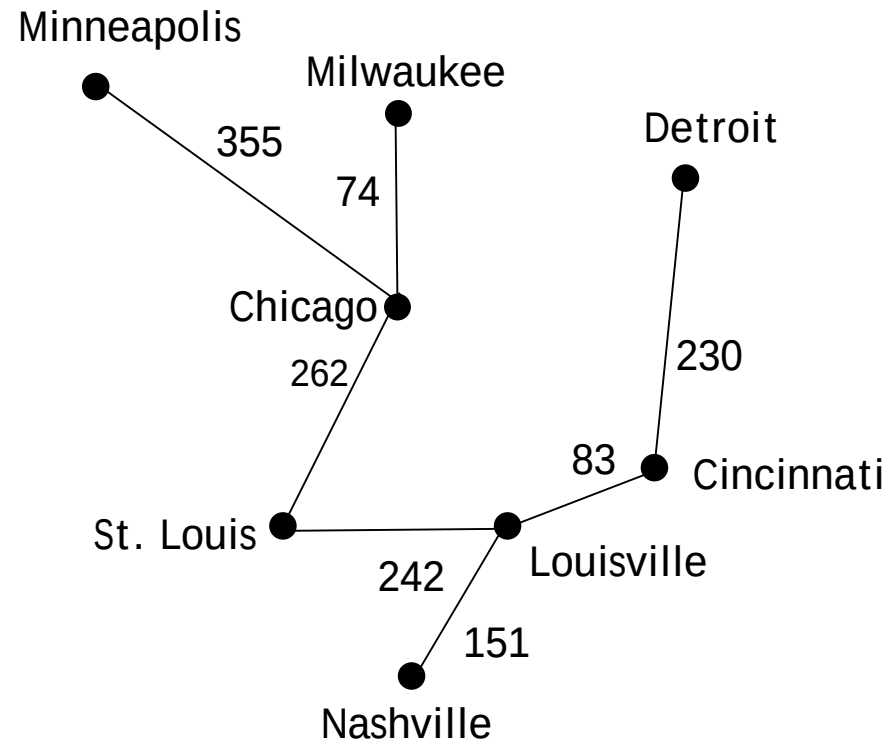
Find the minimal spanning tree for the airline example.

Step 12

Locate the next edge with the least weight :

Minneapolis to Chicago (355)

Add the edge and update $m = 7$.



Application : Minimum Spanning Tree in Communication System (Cont'd)

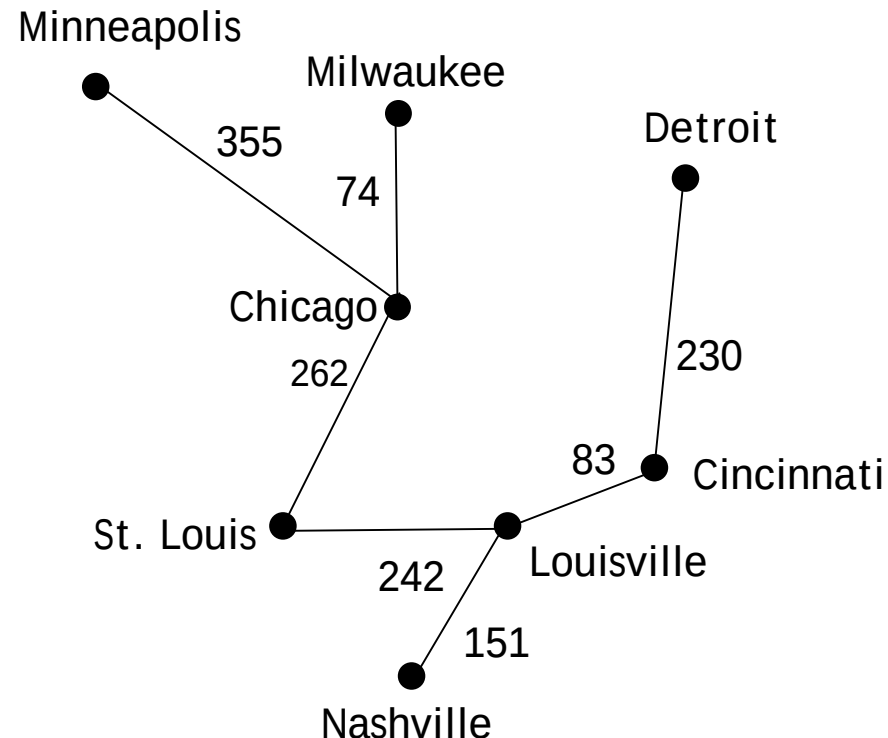
Find the minimal spanning tree for the airline example.

And we have are done.

Verify that every vertex is connected to another vertex and the total mileage is the smallest.

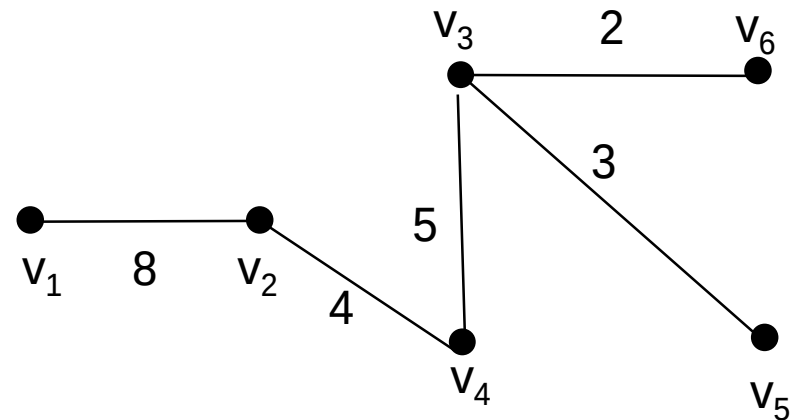
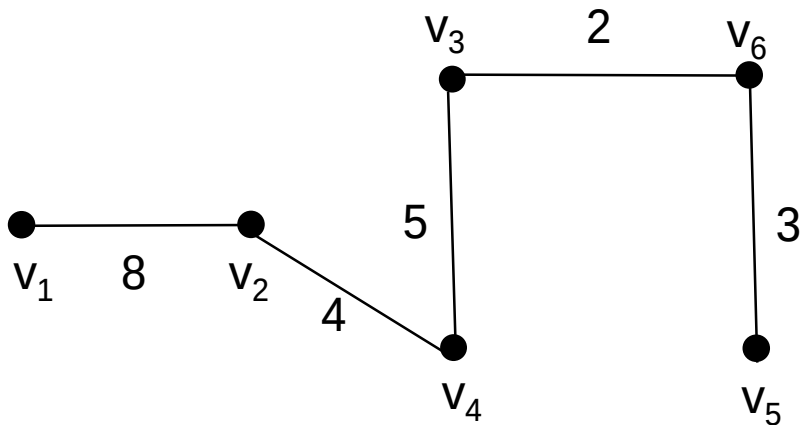
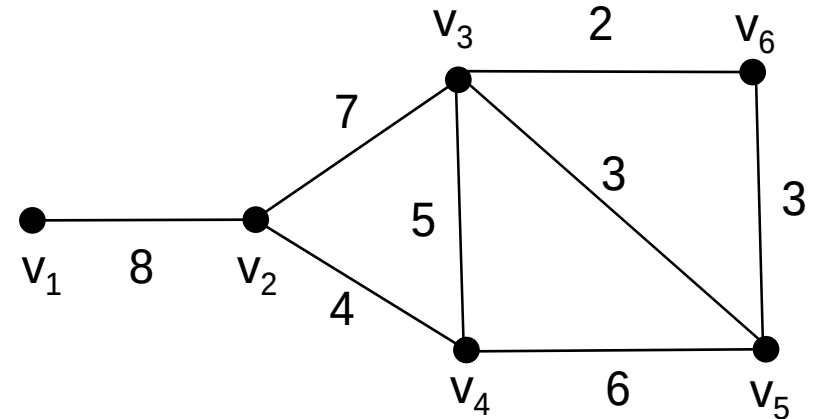
Every minimum spanning tree will have $n - 1$ edges, where n is the number of vertices.

In this case, we have 7 edges.



Exercise (Minimum Spanning Tree)

How many minimum spanning tree can you find? Draw them.



*****END OF TOPIC*****